Mark* VIe PAMC-based CDM

Combustion Dynamics Monitoring for an Existing Mark VI Control

Instruction Guide

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Safety Symbol Legend



Warning

Indicates a procedure, condition, or statement that, if not strictly observed, could result in personal injury or death.



Caution

Indicates a procedure, condition, or statement that, if not strictly observed, could result in damage to or destruction of equipment.



Attention

Indicates a procedure, condition, or statement that should be strictly followed to improve these applications.

Control System Warnings



Warning

To prevent personal injury or damage to equipment, follow all equipment safety procedures, Lockout Tagout (LOTO), and site safety procedures as indicated by Employee Health and Safety (EHS) guidelines.



Warning

This equipment contains a potential hazard of electric shock, burn, or death. Only personnel who are adequately trained and thoroughly familiar with the equipment and the instructions should install, operate, or maintain this equipment.



Warning

Isolation of test equipment from the equipment under test presents potential electrical hazards. If the test equipment cannot be grounded to the equipment under test, the test equipment's case must be shielded to prevent contact by personnel. To minimize hazard of electrical shock or burn, approved grounding practices and procedures must be strictly followed.



Warning

To prevent personal injury or equipment damage caused by equipment malfunction, only adequately trained personnel should modify any programmable machine.



Warning

Always ensure that applicable standards and regulations are followed and only properly certified equipment is used as a critical component of a safety system. Never assume that the Human-machine Interface (HMI) nor the operator will close a safety critical control loop.

Document Updates

Location	Description
ToolboxST Configuration and Download Procedure	Added requirements for the SAMB *Position* field
Location of SAMB Barcode	New section to help TA know how to find the physical location of the SAMB barcode
Collect the Site .m6b File Information	New section to help TA know how to collect the required information from site .m6b file
ToolboxST Configuration and Download Procedure	Updated procedure steps to *set the baseload* on the PAMCs
Point the Mark VIe SDB to the Mark VI SDB	Updated to use the same configuration as listed in the .m6b file (in most cases)
Put the new Mark VIe EDG into the Mark VI SDB	Added the steps to put external Mark VI device to SDB
Tools Configuration Procedures	New Figure
Appendix	New Section

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Acronyms and Abbreviations

CDM	Combustion Dynamics Monitoring
CST	Control Systems Technology is a division of GE Power & Water
EGD	Ethernet Global Data
FFT	Fast Fourier Transform
GEIP	GE Intelligent Platforms is a division of GE Energy Management
HMI	Human-machine Interface
M&C	Measurement & Controls is a division of GE Oil & Gas
MBC	Model-based controls
OSM	Onsite Monitor
PEECC	Packaged Electrical & Electronics Control Compartment
PGS	Power Generation Services is a division of GE Power & Water
SDB	System Database
UCSBH4A	The Mark VIe controller used in this CDM assembly, has 1066 MHz Intel EP80579 processor
UCVx	The Mark VI controller family
UDH	Unit Data Highway (Ethernet network for communications across controllers and HMIs)

Related Documents

Tip *For access to Mark* controls user documents, go to http://techpubs.salem.ge.com and search for other related documents (requires access to GE intranet).*

Doc Number †	Classification	Document Title
GEH-6721_Vol_I	Public	Mark* VIe and Mark VIeS Control System Guide
GEH-6721_Vol_II	Public	Mark* VIe and Mark VIeS Control Systems Volume II: General-purpose Applications
GEH-6721_Vol_III	GE Internal	Mark* VIe and Mark VIeS Control Systems Volume III: For GE Industrial Applications
GEH-6421_Vol_I	GE Internal	Mark* VI Control System Guide, Volume I
GEH-6421_Vol_II	GE Internal	Mark* VI Control System Guide, Volume II
GEH-6808	GE Internal	ControlST* Software Suite How-to Guides
GEH-6700	GE Internal	ToolboxST* User Guide for Mark VIe Control
GIT-S00171	GE Internal	Acoustic Monitor Gateway (AMGateway) Installation Guide
† URL links to download pdf files only work if computer is on a GE network.		

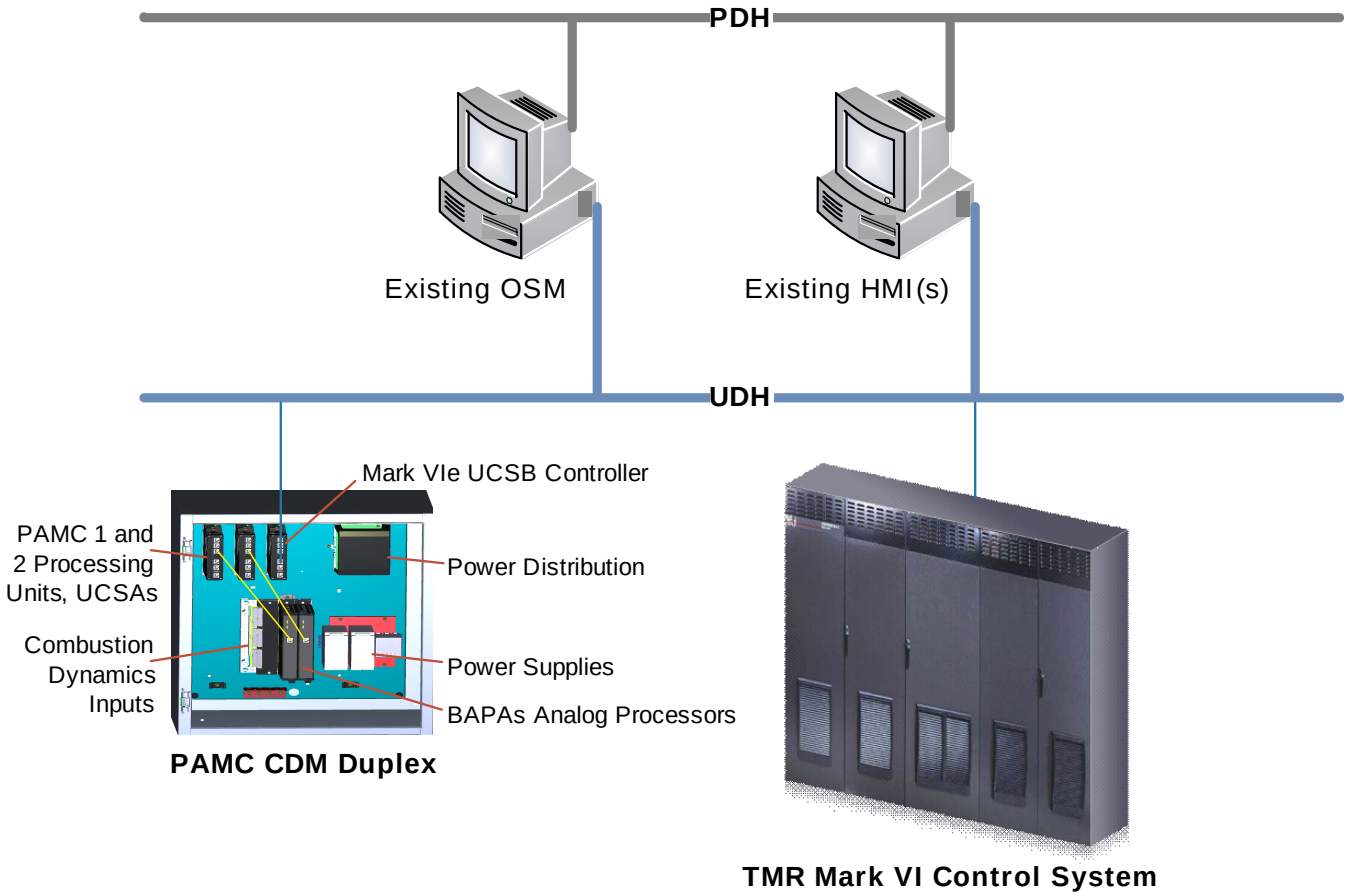
1 Introduction

The Mark* VIe PAMC-based CDM provides the enabling platform for combustion dynamic monitoring of GE gas turbines, used with existing Mark VI controls. This solution replaces the obsolete Mark VI VME-based VAMB boards, enabling GE to continue serving its existing Mark VI fleet with new control application technologies. The following are key motivators for adoption of the PAMC-based CDM:

ETS Auto-tune is also available for the Mark VIe Integrated Control System (ICS).

Refer to the section, CDM Use Cases for more information.

- Product offerings for ETS Auto-Tune, TIL 1931 Dynamics Protection and other applications requiring combustion dynamics monitoring
- Dry Low NOX (DLN) Remote Tuning Service provided to customers with contracts with GE to periodically manually re-tune their gas turbine combustion profiles for either cold weather or warm weather operation
- The PAMC-based CDM is a lower cost, smaller scope alternative to the Mark VI to Mark VIe platform upgrade (as it does not upgrade the Mark VI UCVx controller nor the VPRO protection module)



The following components are included in the PAMC-based solution:

The UDH switch must support a minimum 100 MB/sec speed for the UCSB controller.

- **UCSBH4A Mark VIe controller** connects to the UDH network switch for broadcasting EGD to the existing HMIs, OSMs, and Mark VI controllers.
- **PAMC Acoustic Monitoring module** in either duplex or simplex
  - Duplex for support of combustion dynamics control applications
    or
  - Simplex for support of DLN remote tuning or other combustion dynamics monitoring customer requirements
- Redundant **Phoenix power supplies** for combustion dynamics control applications or a single power supply for the DLN remote tuning product. These power supplies are capable of accepting 110 or 220 V ac, or 125 V dc power
- **JPDH** power distribution board
- Line filter module (ac/dc)

Refer to the Acoustic Monitor Gateway (AMGateway) Installation Guide (GIT-S00171) for more information.

The PAMC-based CDM supports heavy-duty gas turbine applications. ControlST software is used to set up, configure, and download to the PAMC and Mark VIe controller. Mark VI control application changes (legacy Toolbox) are needed to support the new CDM data. An updated AMGateway is needed for full spectrum displays.

1.1 Summary of PAMC-based CDM Upgrade

The PAMC-based CDM is floor-mounted in the PEECC. Its mechanical assembly contains the following components, which are pre-mounted to its back panel:

Separation of the front door and backplate from the main box provides less weight for installation, as it can then be installed in three separate pieces.

- One SAMB terminal board with either one (simplex) or two (duplex) BAPA analog processors
- One or two UCSAH1A modules (PAMC)
- One UCSBH4A Mark VIe controller
- Ethernet cable(s) connecting the UCSA(s), BAPA(s), and UCSB controller
- Power distribution components:
  - One JPDH distribution board
  - One or Two Phoenix power supplies
  - One diode power ORing module (if duplex version)
  - One Corecom power filtering module

The following ControlST software is required for the PAMC-based CDM:

- The ToolboxST* application is used to configure the PAMC and UCSB, and to set up the UDH EGD pages that are broadcast to the existing Mark VI controllers, HMIs, and OSMs.
- Firmware is downloaded to the PAMC and UCSB from the computer that runs the ToolboxST application.

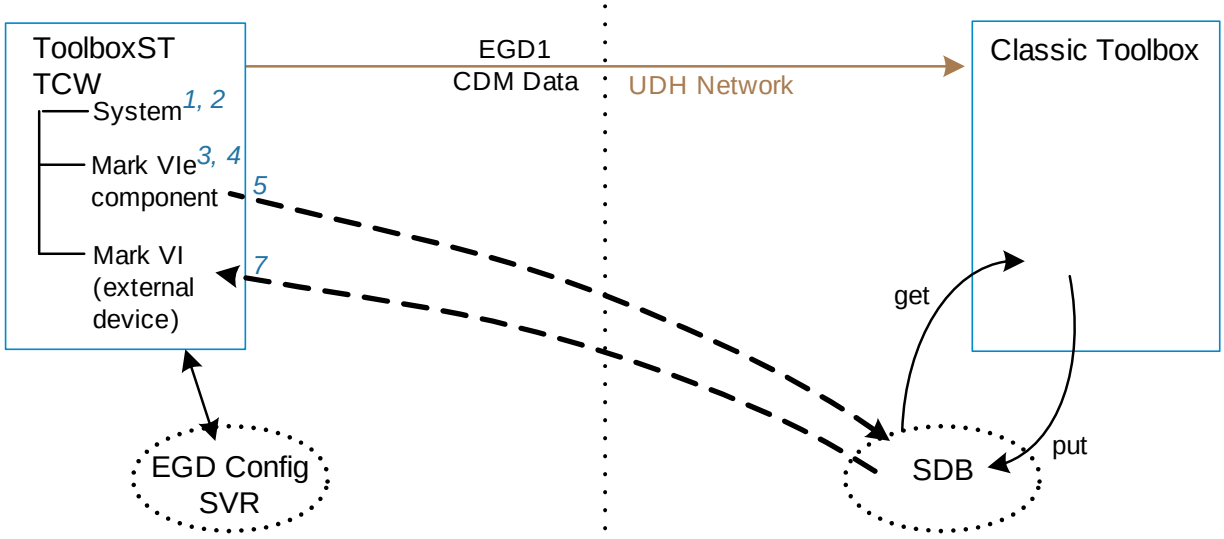
The PAMC-based CDM broadcasts combustion dynamics data from either an Encore charge amplifier or a Gen IV PCB charge amplifier that is connected to the SAMB terminal board. The SAMB fans the analog CDM signals to both BAPA analog processors. The CDM inputs pass through the BAPA bias nulling feature (gain adjustment and anti-aliasing filters) before the analog-to-digital conversion occurs. The CDM digital data is filtered again and down-sampled per the FFT sample-rate needed. The data is then packetized and sent to the UCSA processing unit for the PAMC windowed-FFT and frequency sorting function.

In the duplex version, the PAMC FFT function is performed independently by two separate PAMCs (BAPA + UCSA combinations), based on shared analog data from the SAMB terminal board. PAMC1 and PAMC2 CDM peak magnitudes and corresponding frequencies for each frequency band defined are sent to the Mark VIe controller, which broadcasts these two UDH EGD pages to the existing Mark VI control.

Dual power supply connections are provided on the SAMB terminal board to power both BAPA analog processors and power the GEN IV PCB charge amplifiers with a 24 V dc sourced, 3.6 mA constant-current supply on the same two wires used to receive the CDM signal. The combustion dynamics ride on top of a 12 V dc nominal bias voltage of the charge amplifier. The JPDH power distribution board provides branch protection for the PAMC module and the Mark VIe controller.

2 Tools Configuration Procedures

The following diagram outlines the high-level steps required to configure the Mark VIe PAMC-based CDM and interface to the existing Mark VI control.



ToolboxST Configuration Steps

1. Configure the IP Address of the master SDB .
2. Set the path to the SDB.
3. Set the NTP server and EGD network name
4. Configure the PAMC-based CDM.
5. Put the CDM EGD pages to the SDB.
7. Get EGD from the SDB.

Toolbox Changes

6. Update the Mark VI controller application logic .

2.1 Minimum ControlST* Version Selection

Prior to installation of the ControlST software suite, it is important to have a full understanding of this tools package, especially the ToolboxST application. Take some time reading the following two documents, and be especially aware of the minimum computer hardware and operating system requirements for the particular version of ControlST software that will be installed for the customer.

The following hyperlinks will only work from the GE Internal Network.

- GEH-6700, ToolboxST User Guide for Mark VIe Control
- GEI-100694, ControlST Software Suite Upgrade Instruction Guide

2.1.1 Mark VIe PAMC-based CDM Without Full Spectrum Support (Not Integrated with AMGateway)

➢ **To determine HMI requirements, follow these decision-making steps**

1. If the site system already has ControlST V04.07 or higher installed, then use the existing toolset.

2. If the site system does not have ControlST V04.07 or higher, then check the following:

*The Mark VIe to Mark VI configuration transfer requires *.ALB files (Generic Network Device) to define the Mark VIe EGD pages to the Mark VI control.*

a. If the site system is ControlST-based, then check:

i. If the customer will permit a site upgrade, then install ControlST V04.07 or higher.

ii. If the customer will not permit a site upgrade, then install ControlST V05.02 or higher with an IC647CSTRENEW, Premium Renewables Software License Key on a separate computer.

b. If the site system is eTCSS-based Toolbox 11, then check:

i. If the site HMI meets the minimum computer requirements, then ToolboxST V05.02 or later can be installed on the site HMI and used to configure the Mark VIe CDM. This requires an IC647CSTRENEW, Premium Renewables Software License Key.

ii. If the site HMI does not meet minimum computer requirements, then ToolboxST V05.02 or later with an IC647CSTRENEW, Premium Renewables Software License Key must be installed on a separate computer.

c. If the site system is eTCSS-based Toolbox 10 or older, then ToolboxST V05.02 or later with an IC674CSTRENEW, Premium Renewables Software License Key must be installed on a separate computer.

2.1.2 Mark VIe PAMC-based CDM with Full Spectrum Support (Integrated with AMGateway)

➢ **To determine HMI requirements, follow these decision-making steps**

1. If the site system is ControlST-based, then check the following:

a. If the customer will permit a site upgrade and the HMI meets the minimum computer requirements, then install ControlST V05.04 or later with the IC647CST, Full ControlST Software License Key.

b. If the customer will not permit a site upgrade, then the customer can only run a Mark VIe PAMC-based CDM without full spectrum support.

c. If the full spectrum information is to support the Remote DLN Tuning service only, then install ControlST V05.04 or higher with an IC647CST, Full ControlST Software License Key on the site's OSM only.

2. If the site system is eTCSS-based Toolbox 11, then check the following:

a. If the site HMI meets the minimum computer requirements, then ToolboxST V05.02 or later can be installed on the site HMI and used to configure the Mark VIe CDM. The IC647CSTRENEW, Premium Renewables Software License Key is required.

b. If the site HMI does not meet the minimum computer requirements, then ToolboxST V05.02 or later with IC647CSTRENEW Premium Renewables Software License Key must be installed on a separate computer.

c. Install the eTCSS version of the AMGateway V02.06 or later on the HMI computer.

3. If the site system is eTCSS-based Toolbox 10 or older, then

a. If the site HMI is running TCI 1.6 then install eTCSS V02.06 or later of AMGateway.

b. If the site HMI is running TCI 1.5 or older then complete the following:

i. Upgrade to TCI 1.6

ii. Install eTCSS V02.06 or later of AMGateway.

c. Install ToolboxST V05.02 or later with IC64CSTRENEW Premium Renewables Software License Key on a separate computer.

d. Mark VIe to Mark VI EGD configuration transfer requires the pre-defined .alb file (Generic Network Device) be put onto the legacy Toolbox computer. This file is distributed and maintained by the GE PGS group.

2.2 Configuration and Download of Mark VIe PAMC-based CDM

2.2.1 Prerequisites

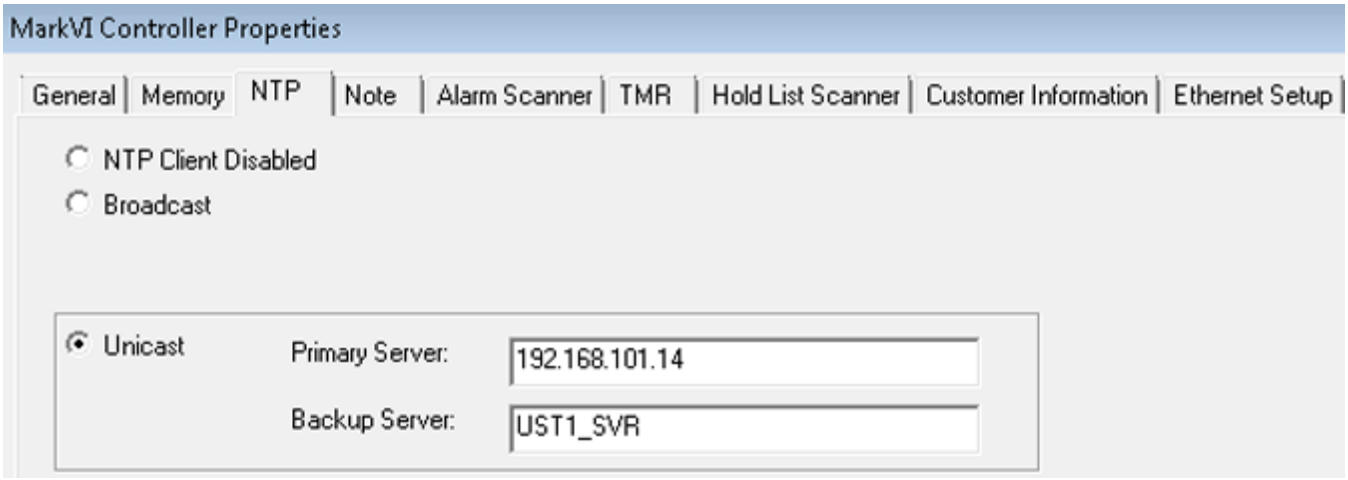
- The correct versions of ControlST software suite, legacy Toolbox application, and AMGateway have been installed at site by using the decision-making steps in the section, *Minimum ControlST Version Selection*
- The ToolboxST .tcw system file with support files are included in a subfolder named, *CDM* on the computer that runs ToolboxST application.
- The required information from site .m6b file
- If using the GE laptop for PAMC-based CDM, be sure to have the standard HMI passwords for access.

2.2.2 Collect the Site .m6b File Information

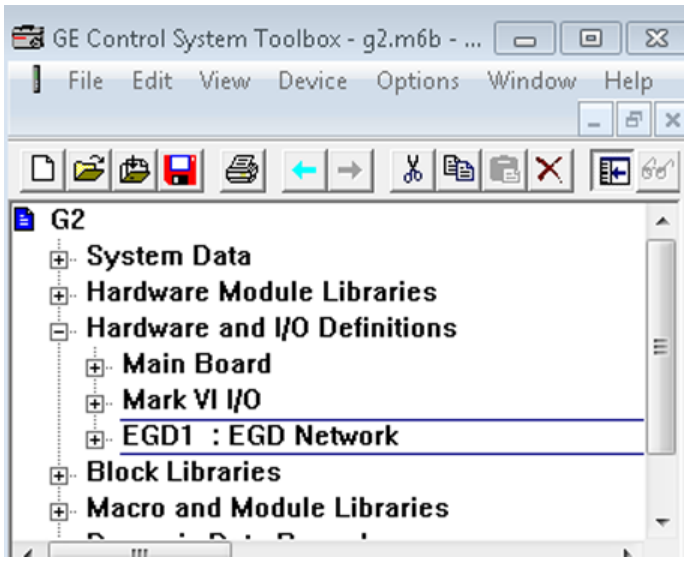
Use the following procedure to collect the site's NTP server, SDB server, and PDH address information from the Mark VI .m6b file.

➢ **To collect the site-specific configuration**

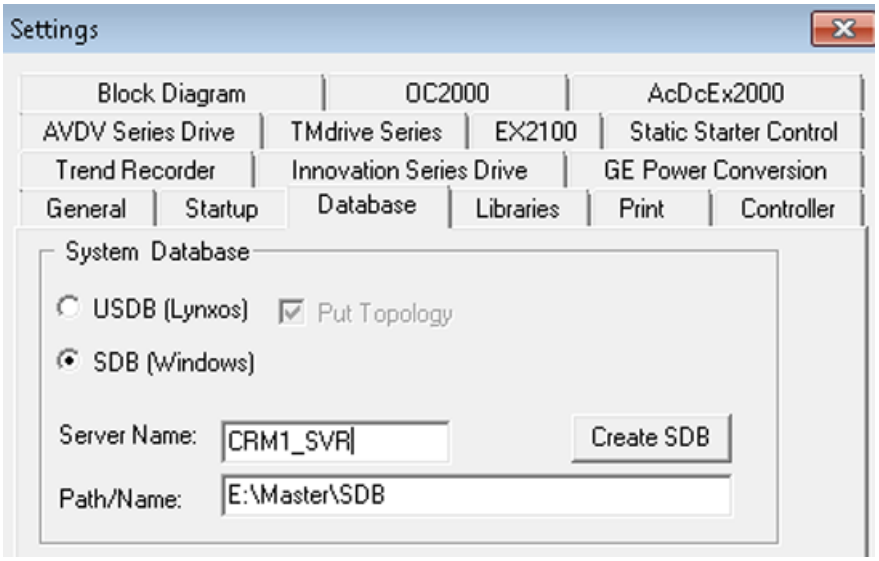
1. Double-click the .m6b file to open it in the legacy Toolbox.
2. Elevate privileges to a minimum level 2.
3. From the NTP tab, make note of the site-specific NTP settings. This screen shot is only an example.



4. Make note of the site-specific **EGD Network** name (to be used as the UDH network in the .tcw). In this screen shot it is EGD1, but this is only an example.



5. From the Options, Settings, **Database** tab make note of the SDB server information.



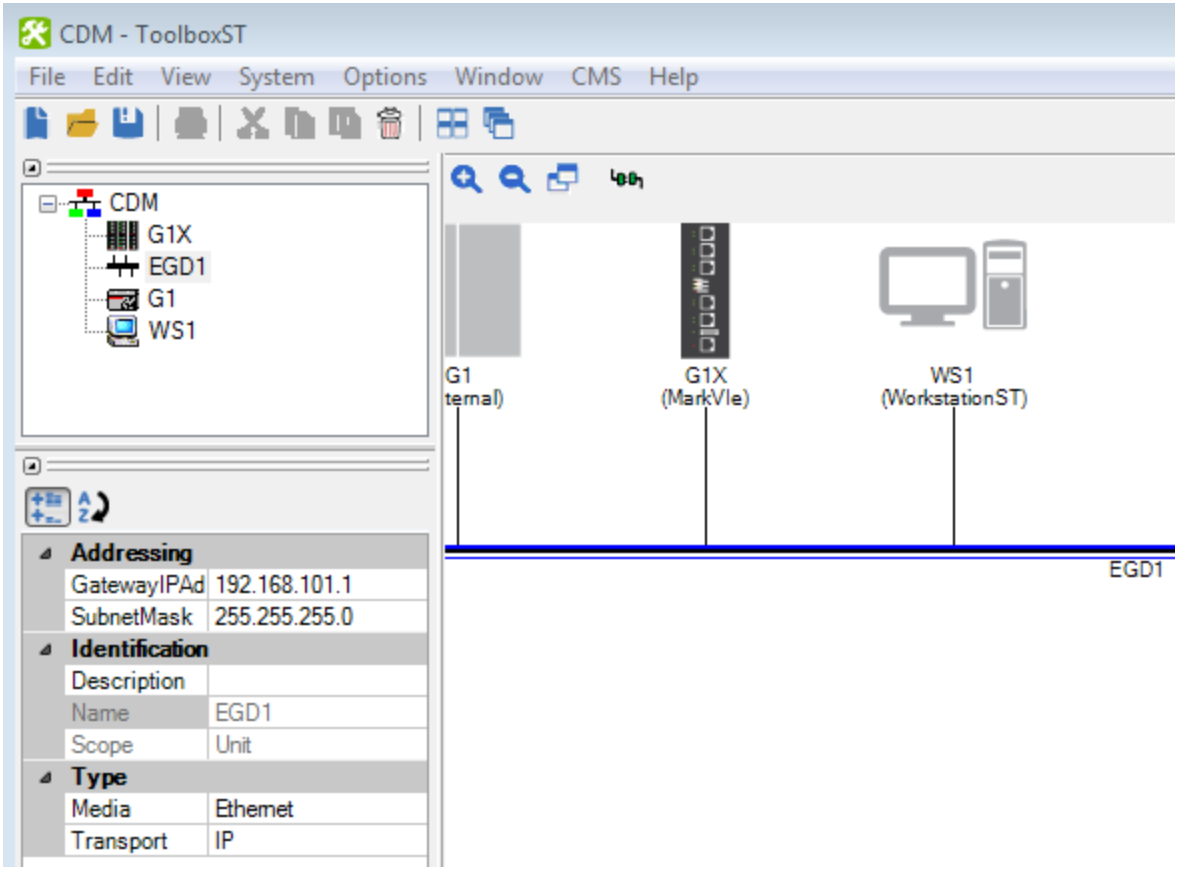
Note If the computer that runs ToolboxST is on the same PDH as the master HMI for Mark VI control, then use the Server Name as displayed from the .m6b file (most common). If not both are on the PDH, then the ToolboxST configuration may need to use the UDH address to get to the SDB.

2.2.3 ToolboxST Configuration and Download Procedure

1. Create a folder named *Projects* on the computer that has the ToolboxST application installed.
2. Copy the subfolder named *CDM* into the *Projects* folder. The *CDM* folder contents are maintained and distributed by GE PGS group.
3. Open the CDM.tcw file located in the *CDM* subfolder.

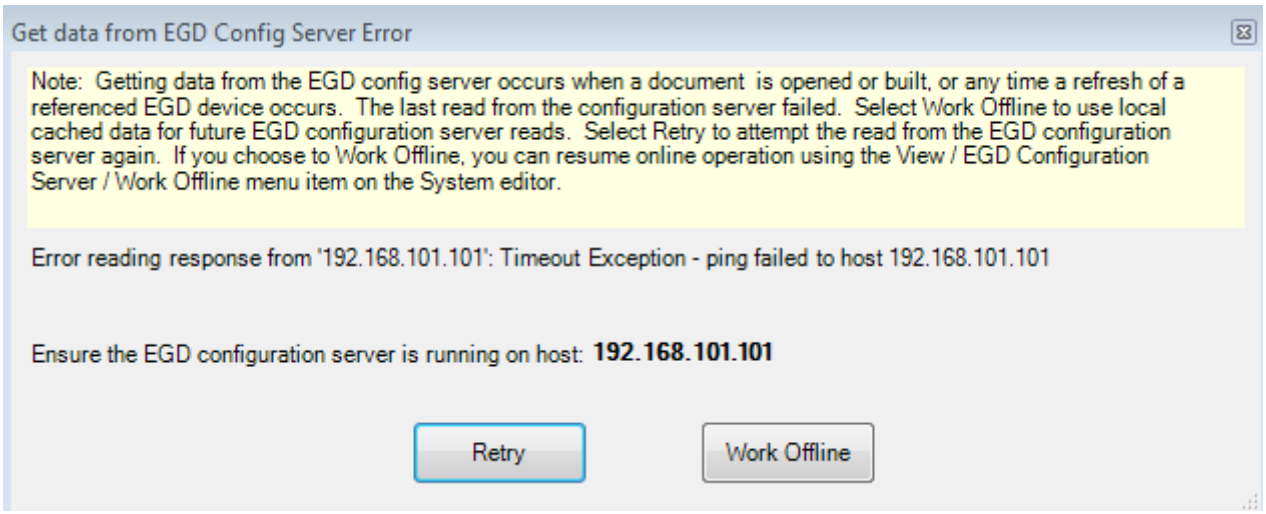
The CDM configuration file opens in the System Editor with the following components displayed:

- G1X – Mark VIe PAMC-based CDM associated with the G1 Mark VI Gas Turbine control
- EGD1 – EGD or UDH network
- G1 – Mark VI Gas Turbine Control
- WS1 – Mark VIe WorkstationST

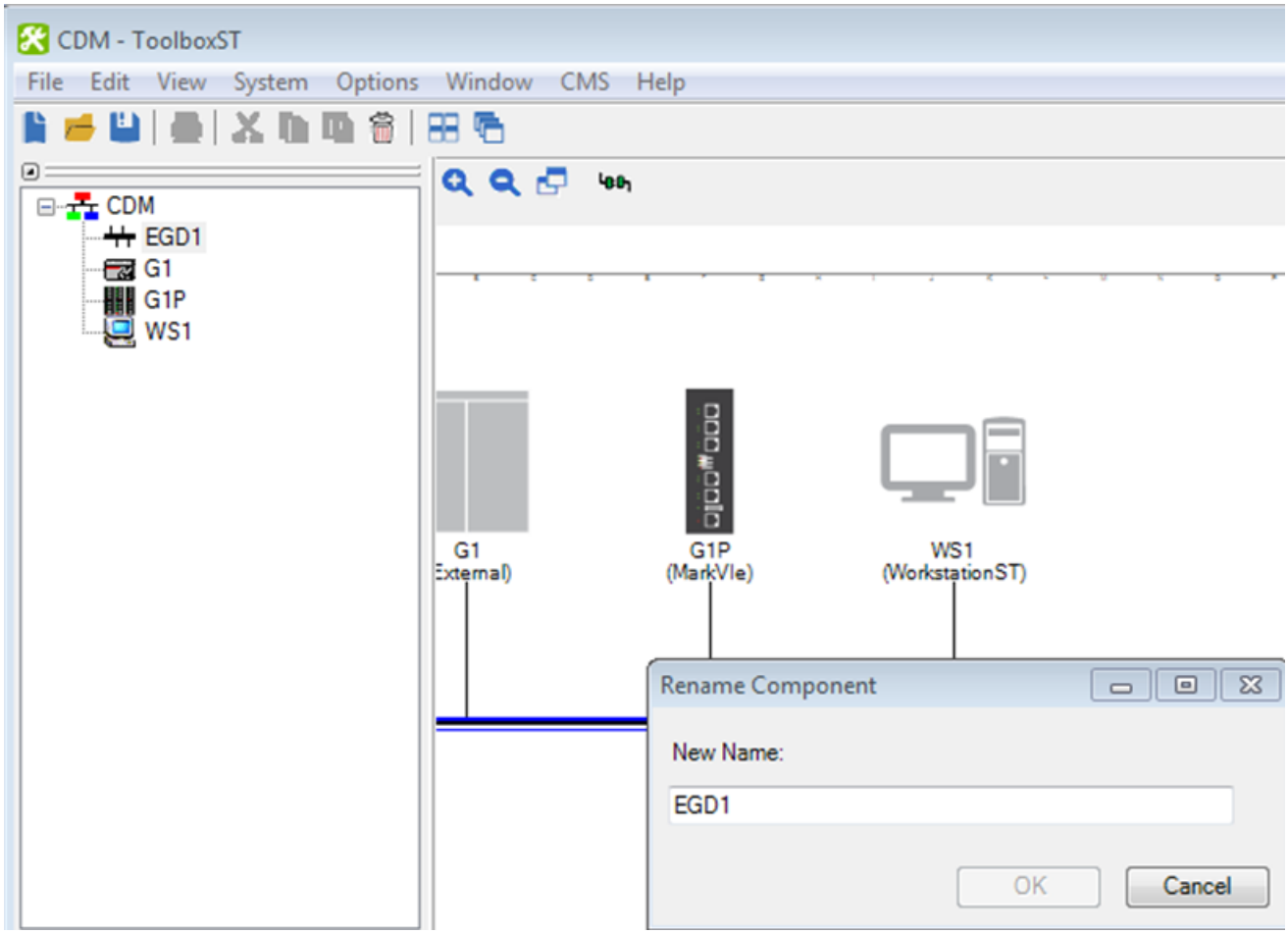


This error will be corrected later.

4. An **EGD Config Server Error** similar to the following message displays. Click **Work Offline**.



5. Change the default UDH name to match the name used at the customer's Mark VI site.



From the System Editor, Tree View, right-click the UDH network (in this example EGD1), select **Rename**, and enter the customer's Mark VI site UDH name.

6. Change EGD or UDH subnet mask to match the subnet mask at the Mark VI site. From the System Editor, Tree View, right-click the UDH network, select the current **Subnet Mask** value, and enter the new value.

7. Change Mark VIe PAMC-based CDM name to the following: <Turbine name>P. For example, if the gas turbine control name is G1, then the associated PAMC-based CDM name will be G1P. From the System Editor Tree View, right-click the Mark VIe controller icon, select **Rename**, enter the new name.

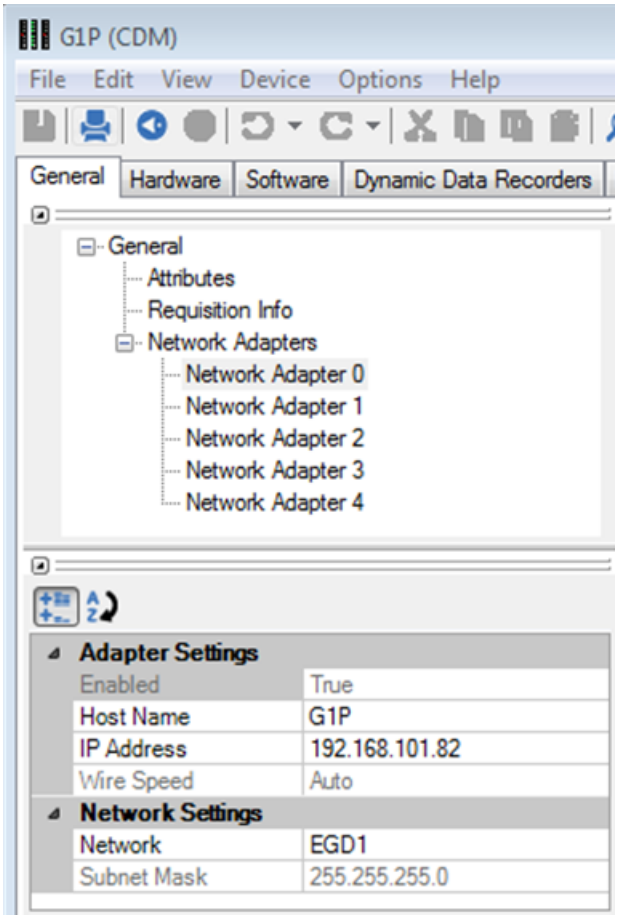
8. Verify that the computer running the ToolboxST application has a properly mapped M drive letter that points to the location of the SDB on the PDH (as indicated in the .m6b file). If not, map the network drive to the correct share folder.

9. From the ToolboxST System Properties, set the **System Database** as follows:

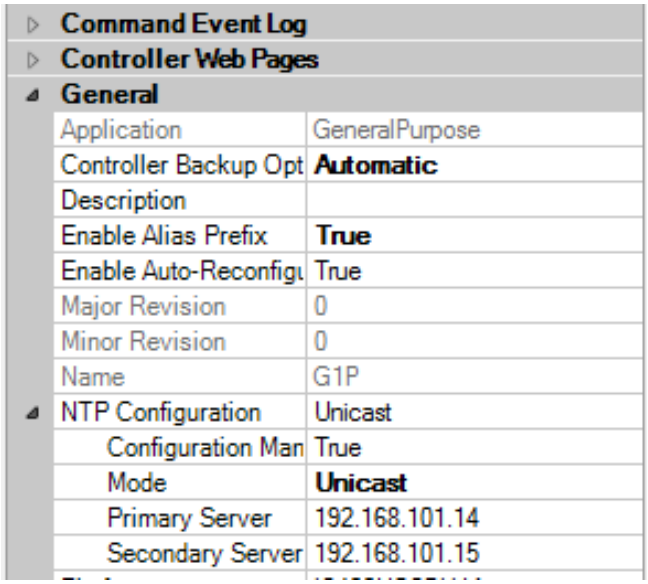
SDB Host Name	If the computer that runs ToolboxST is on the same PDH as the master HMI for Mark VI control, then use the **Server Name** as displayed from the .m6b file (most common). If not both are on the PDH, then the ToolboxST configuration may need to use the UDH address to get to the SDB.
SDB Host Path	The same as listed in the .m6b file

10. Double-clicking the Mark VIe controller icon to open the Component Editor.

11. From the Component Editor, General tab, Property Editor, select **Network Adapter 0**. Verify that the **Host Name**, **IP Address**, and **Network** (name) are configured correctly to match the site's existing EGD Network (found in .m6b file) or change if needed in the .tcw file.



12. Configure the NTP server for the Mark VIe controller to use the same NTP server used in the Mark VI control. The following is an example of this configuration.



13. From the Component Editor tool bar, click **Build**.

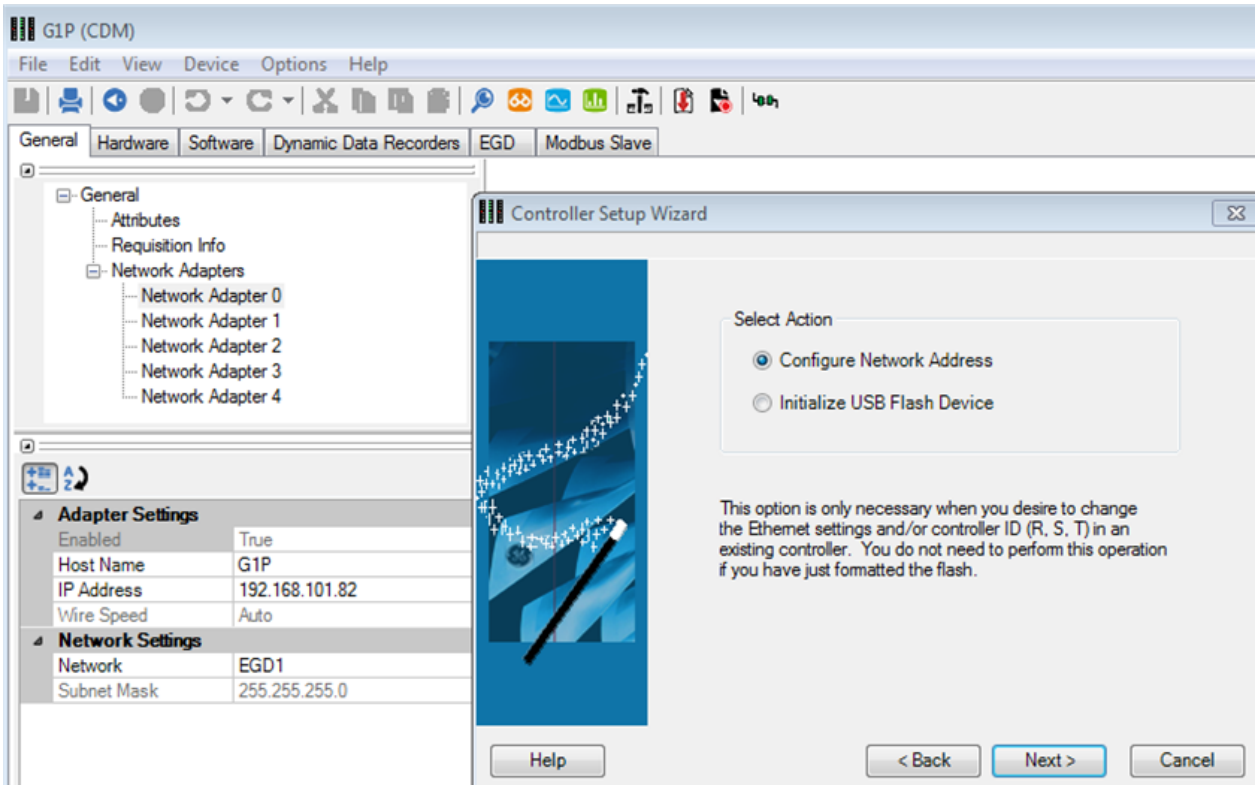
14. Power up the Mark VIe PAMC-based CDM.

15. Set up the Mark VIe controller IP address.

    a. If using the preferred method with unencrypted USB memory stick, refer to *ControlST Software Suite How-to Guides* (GEH-6808), the section *How to Setup and Download to Mark VIe and Mark VIeS UCSB Controllers and I/O Modules* for details instructions. Skip to step 16.

    b. If using the serial cable to COM1 port method, connect a serial cable between the computer that is running ToolboxST application and the UCSB controller COM1 port. This cable (GE part #342A4944P1) must be used for the COM1 port of the UCSB. For computers that do not have a serial port, use a USB to serial port adapter.

i. From the Component Editor **Device** menu, select **Download**, **Controller Setup**, click **Next**, select **Configure Network Address**, and complete the **Controller Setup Wizard**.



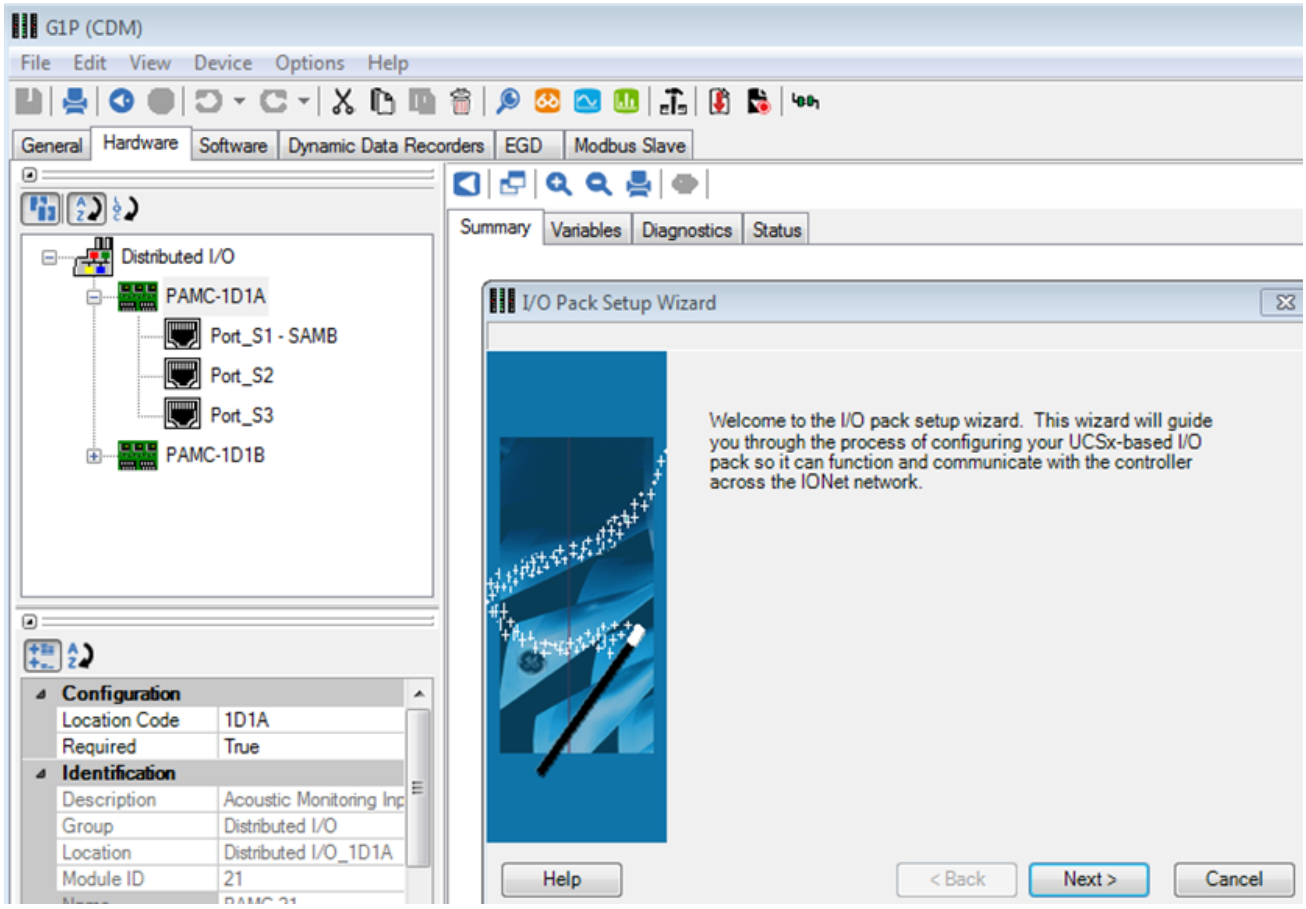
ii. Cycle the power on the Mark VIe PAMC-based CDM.

16. Connect the Mark VIe UCSB controller ENET1 port to the UDH network switch.
17. Connect the computer with ToolboxST application to UDH network switch.
18. From the Component Editor tool bar, click **Download**, and complete the wizard.
19. Wait and then verify that the UCSB controller Boot LED is not lit.
20. Setup the PAMC UCSA(s) initial baseload.

Details for this procedure are documented in the ToolboxST User Guide for Mark VIe Control (GEH-6700), the section Configure PAMC I/O Module.

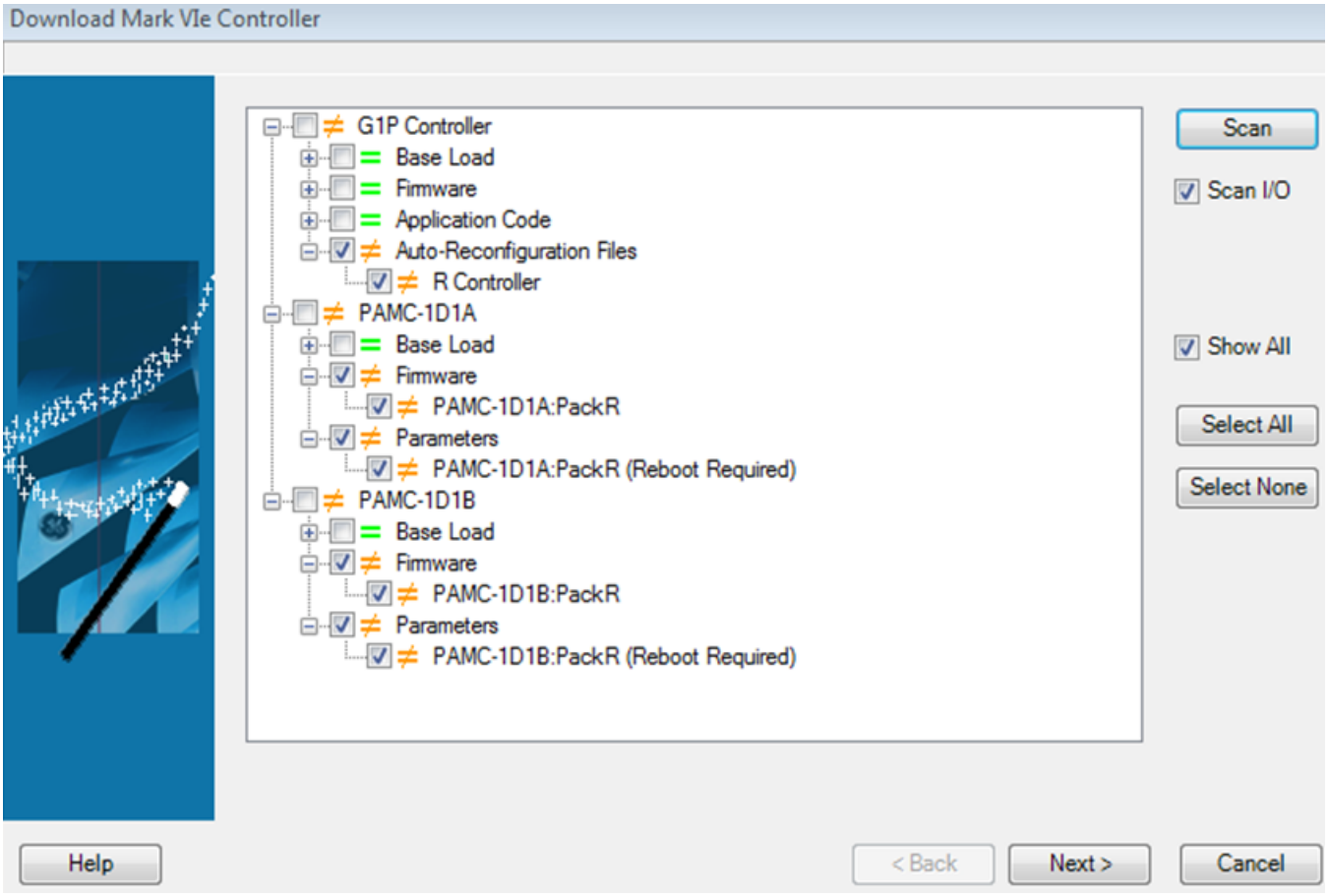
a. Connect a serial cable between the computer that is running ToolboxST application and the UCSA COM1 port. This cable (GE part #342A4944P1) must be used for the COM1 port of the UCSA. For computers that do not have a serial port, use a USB to serial port adapter.

b. From the ToolboxST application start the **I/O Pack Setup Wizard**. From the Component Editor, Hardware tab, Tree View, right-click the PAMC and select Setup.

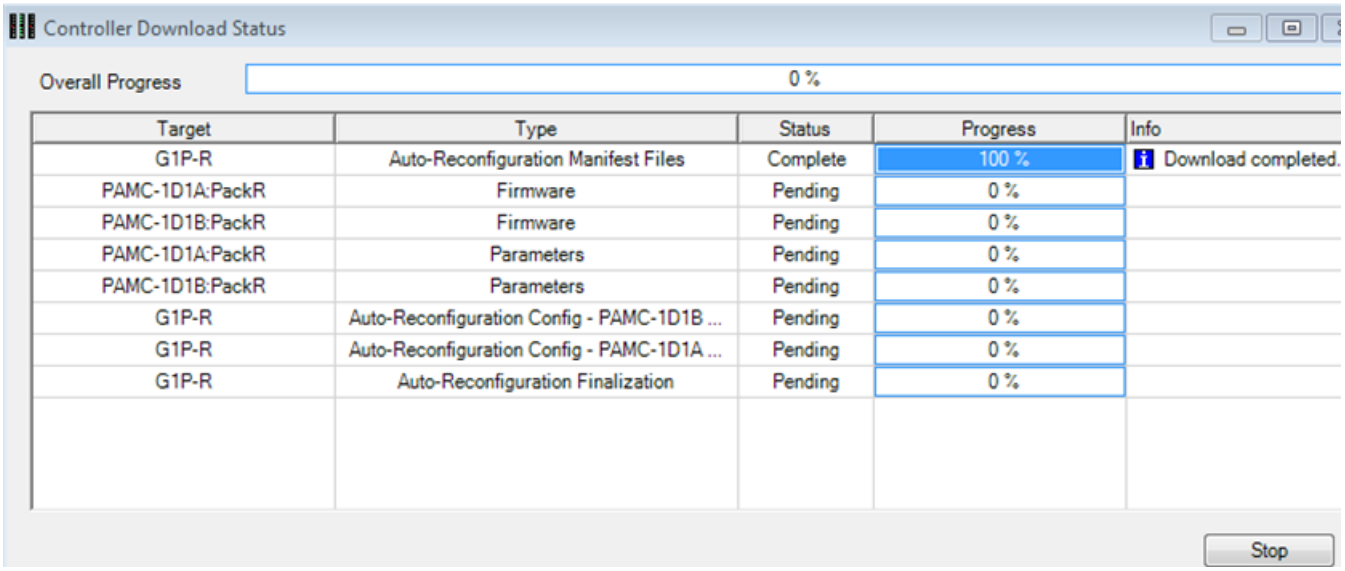


c. If a duplex unit, repeat these steps to setup the initial baseload for the second PAMC.

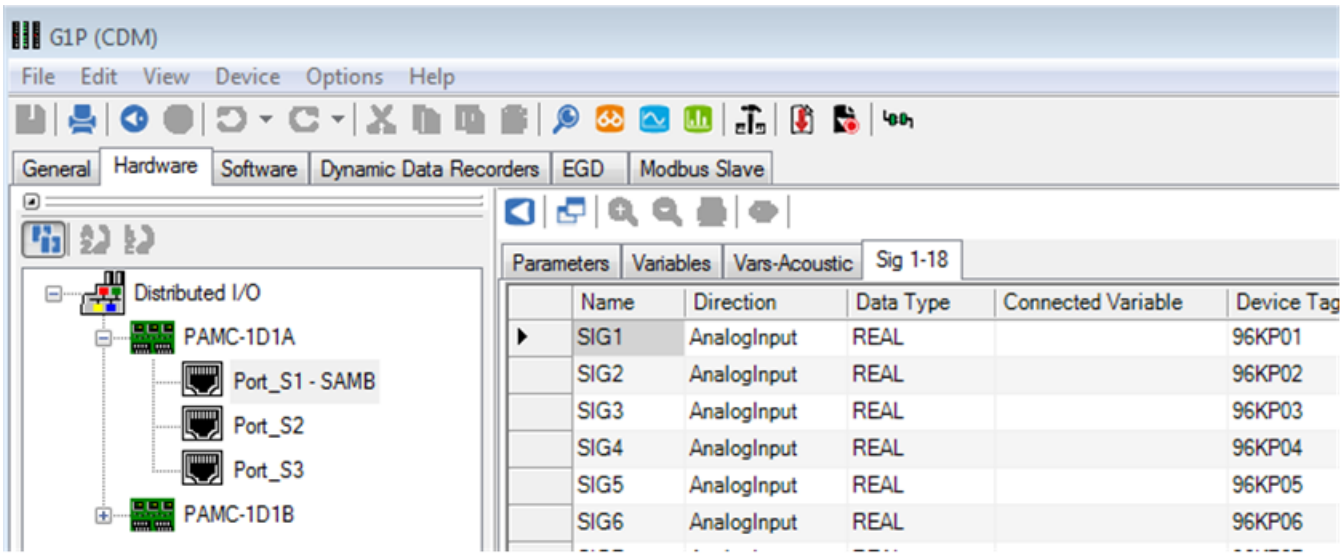
21. From the ToolboxST application, perform a build and download of the configuration (controller and PAMCs).



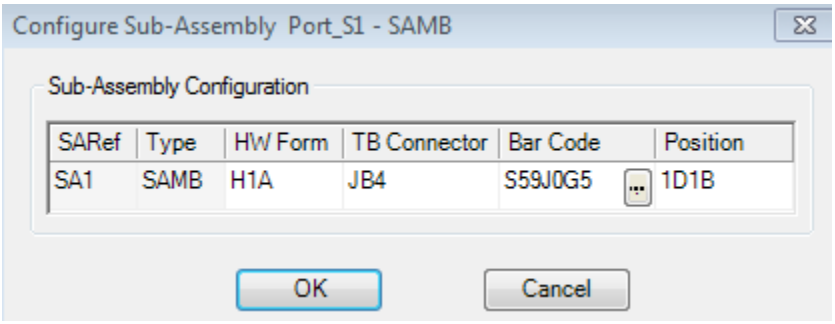
The following window shows an example of the controller Download Status for the PAMC-1D1x where x = A or B.



22. Correct the Acquisition Board barcode mismatch on HSSL link 1 by selecting the Port_S1 – SAMB.



a. From the Tree View, right-click the SAMB and select **Modify**.

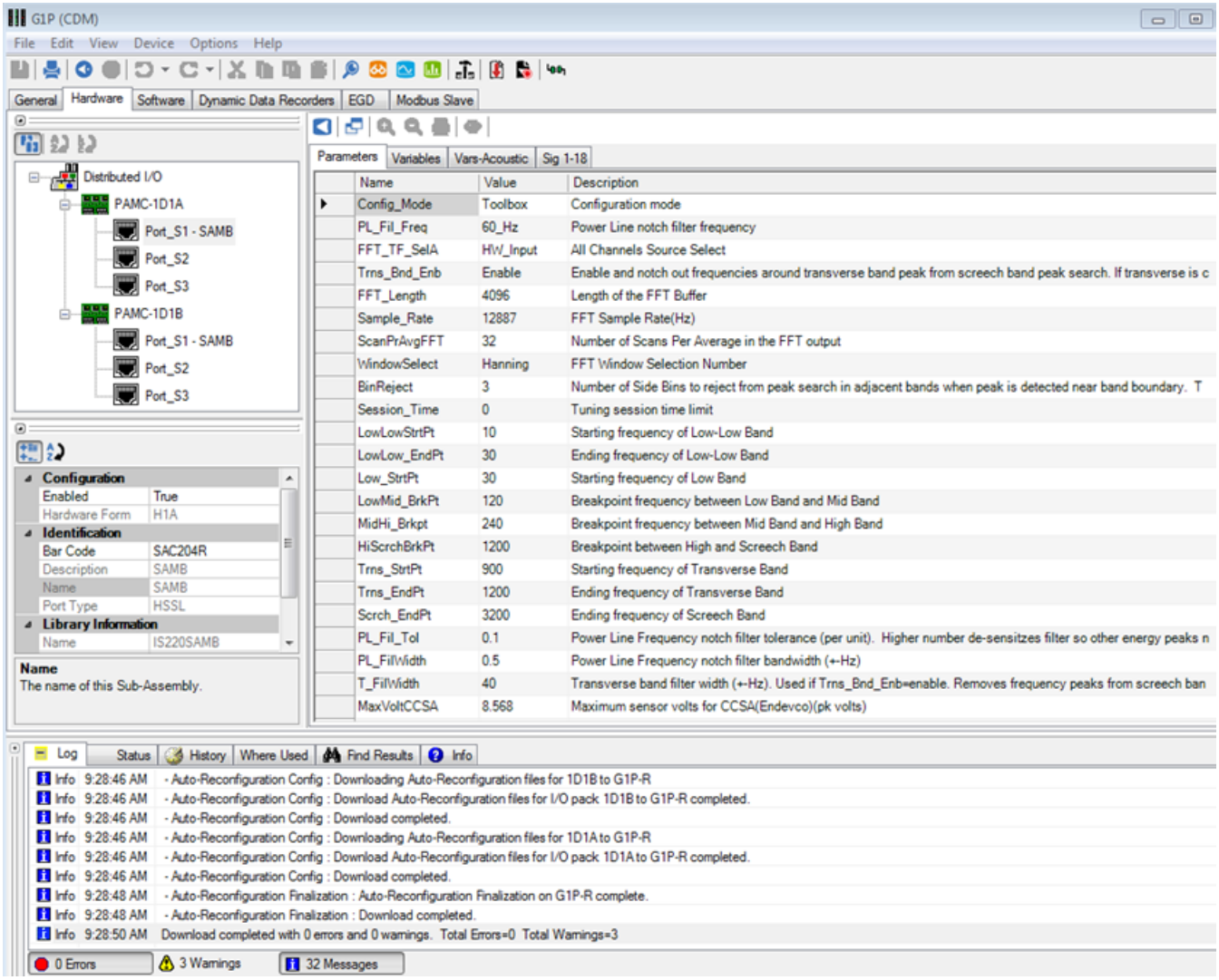


b. Verify that the **Position** contains at least one Alpha character, and only Alpha and Numeric characters.

c. Select the ellipsis under **Bar Code**, select **YES**. ToolboxST will read the new barcode and automatically fill in the new value. Click **OK**.

d. If ToolboxST does not recognize the SAMB barcode, use the following procedure to *locate the SAMB barcode* and then type the barcode into the **Bar Code** field.

23. Click **Build**, and then click **Download**.



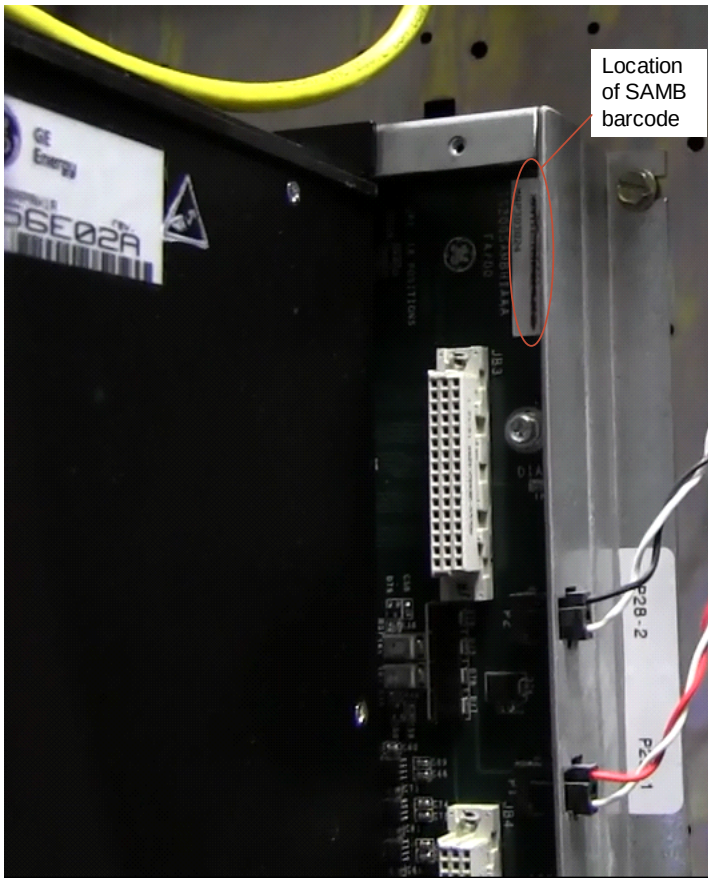
24. From the Info View, verify that the PAMC-based CDM has **0 Errors**.

2.2.4 Location of SAMB Barcode

Use this procedure to locate the SAMB barcode only if there is an issue with ToolboxST application not able to detect the barcode.

➢ **To get the SAMB barcode**

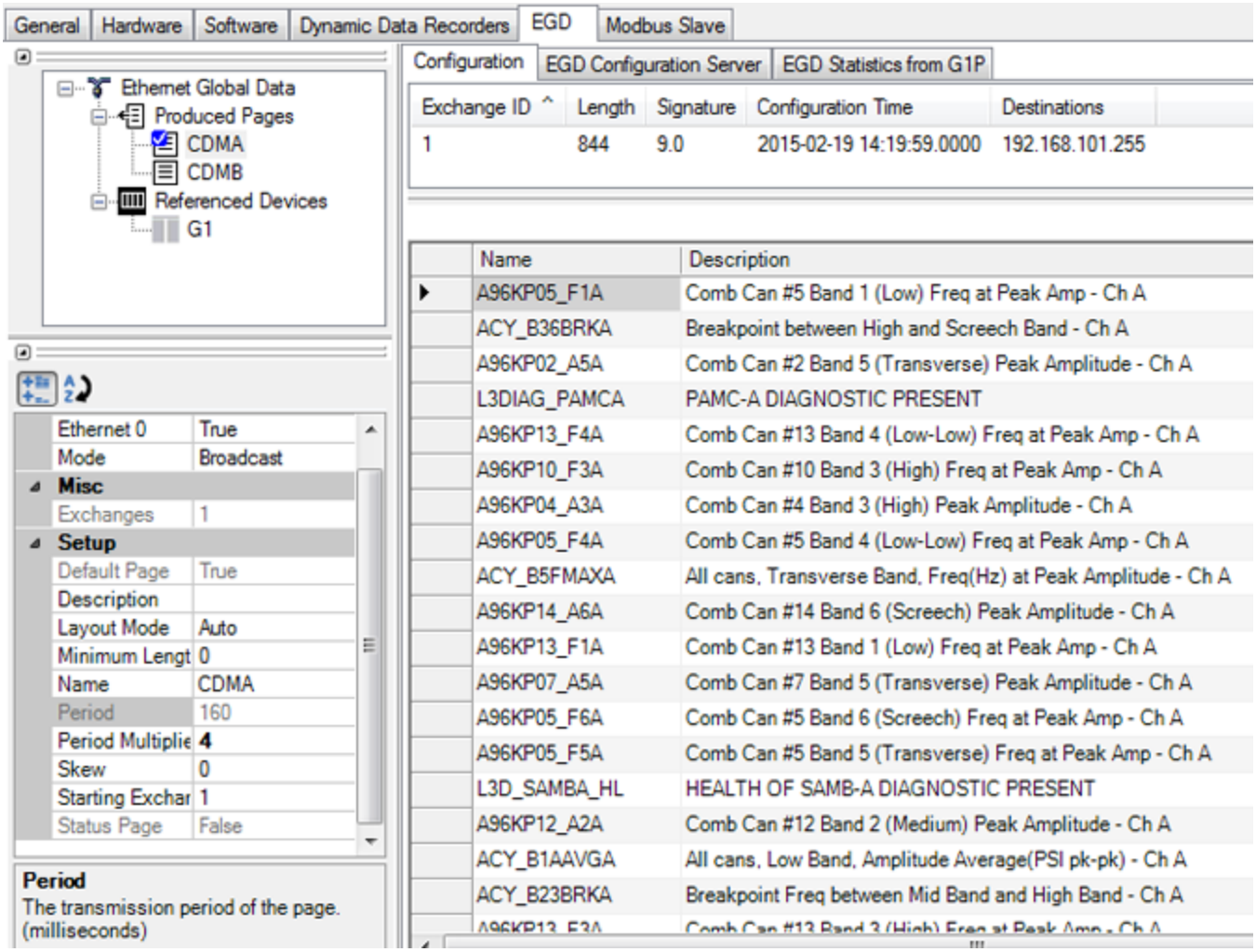
1. Remove the BAPA from the right most slot of the SAMB (or remove the slot cover if simplex).
2. Look at the upper right most corner of the SAMB to find the barcode.



2.2.5 Verify PAMC EGD Pages

➢ **To verify that the PAMC EGD Pages are correct**

1. If the site is running Toolbox 10 or lower, then the site will need *.ALB files (Generic Network Device) that define the layout of the Mark VIe EGD pages. GE PGS group maintains and distributes these files. If the site is running Toolbox 11 or higher, it already has an SDB that both subsystems can use.
2. From the ToolboxST Component Editor, select the **EGD** tab. Verify that both the CDMA and CDMB EGD pages appear under the **Produced Pages** section.



3. Verify that the two EGD exchanges CDMA and CDMB are updating at the appropriate rate based on the PAMC configuration parameter, **FFT_Length**.

FFT_Length	Period	Period Multiplier
1024	40	1
2048	80	2
4096	160	4
8192	320	8

4. Select CDMA (the producer side for the CDM). Go Online and check for Live values.

2.3 Reconcile the Mark VI SDB and the Mark VIe EGD Configuration Server

These steps allow the Mark VI control to receive the data from the Mark VIe control, process it, and expose the CDM data through an EGD page to the other devices (HMIs, OSMs, Historians, and so forth) that require the data.

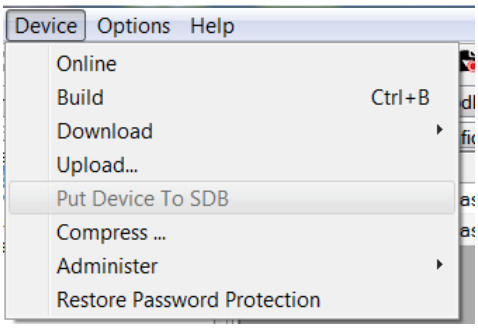
➢ **To put the Mark VIe configuration into the site's Mark VI SDB.**

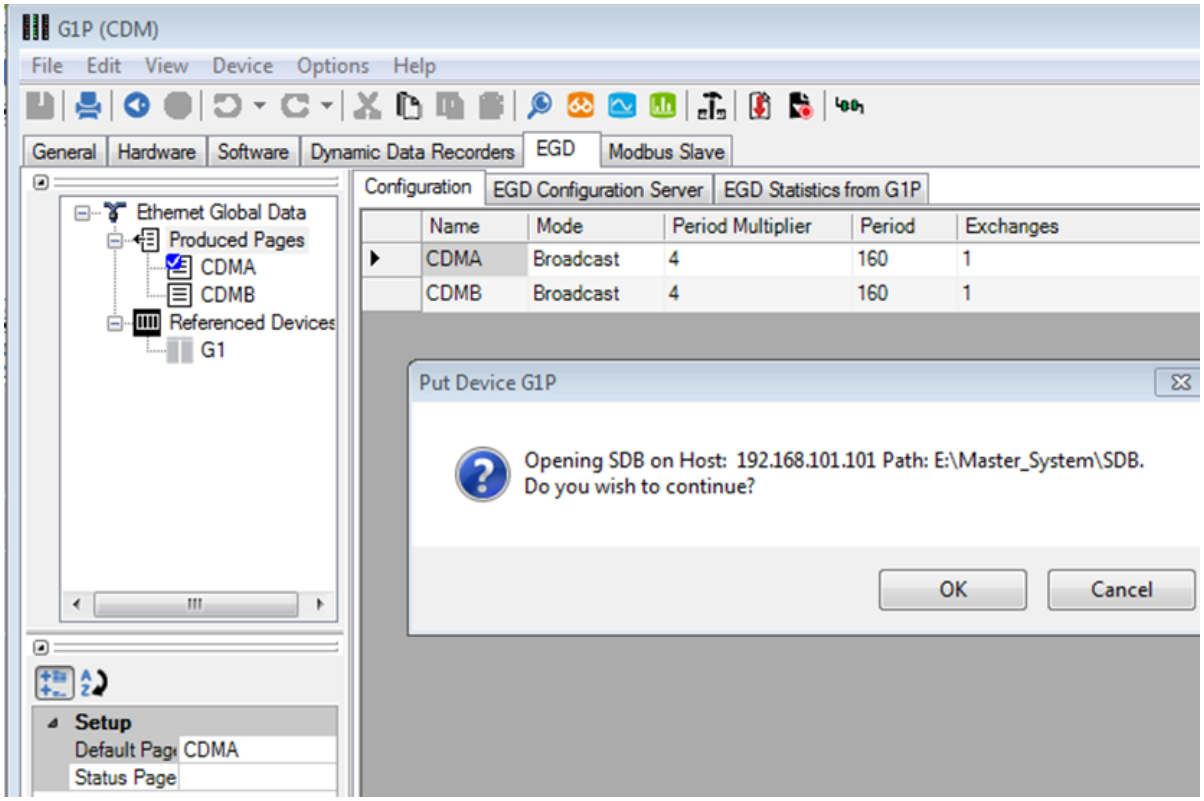


Caution

Never put a WorkstationST Device to the SDB.

1. From the Mark VIe controller Component Editor **Device** menu, select **Put Device to SDB**.



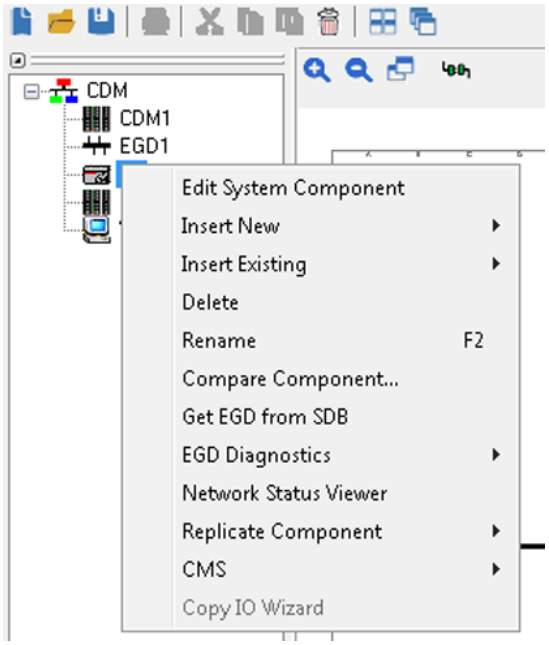


2. Click **OK**.

Note Based on application needs, the Mark VI controller logic is updated during this step of the process (prior to performing a get EDG from SDB to the Mark VI external device (ToolboxST component).

➢ **To get EDG from SDB**

1. From the ToolboxST System Editor Tree Menu, right-click the Mark VI External Device.



2. Select **Get EGD from SDB**.

2.4 Configuration of AMGateway, If Needed

Refer to the Acoustic Monitoring (AM) Gateway Instruction Guide, GEI-100718 for the details on installing the AMGateway.

2.5 Mark VIe PAMC Parameters

Refer to the PAMC Acoustic Monitoring Input Module chapter in GEH-6721 Vol. III for a description on how to configure the charge amplifier and dynamic pressure gains. Also, configure the frequency band boundaries per the gas turbine combustor characteristics.

3 Hardware Installation

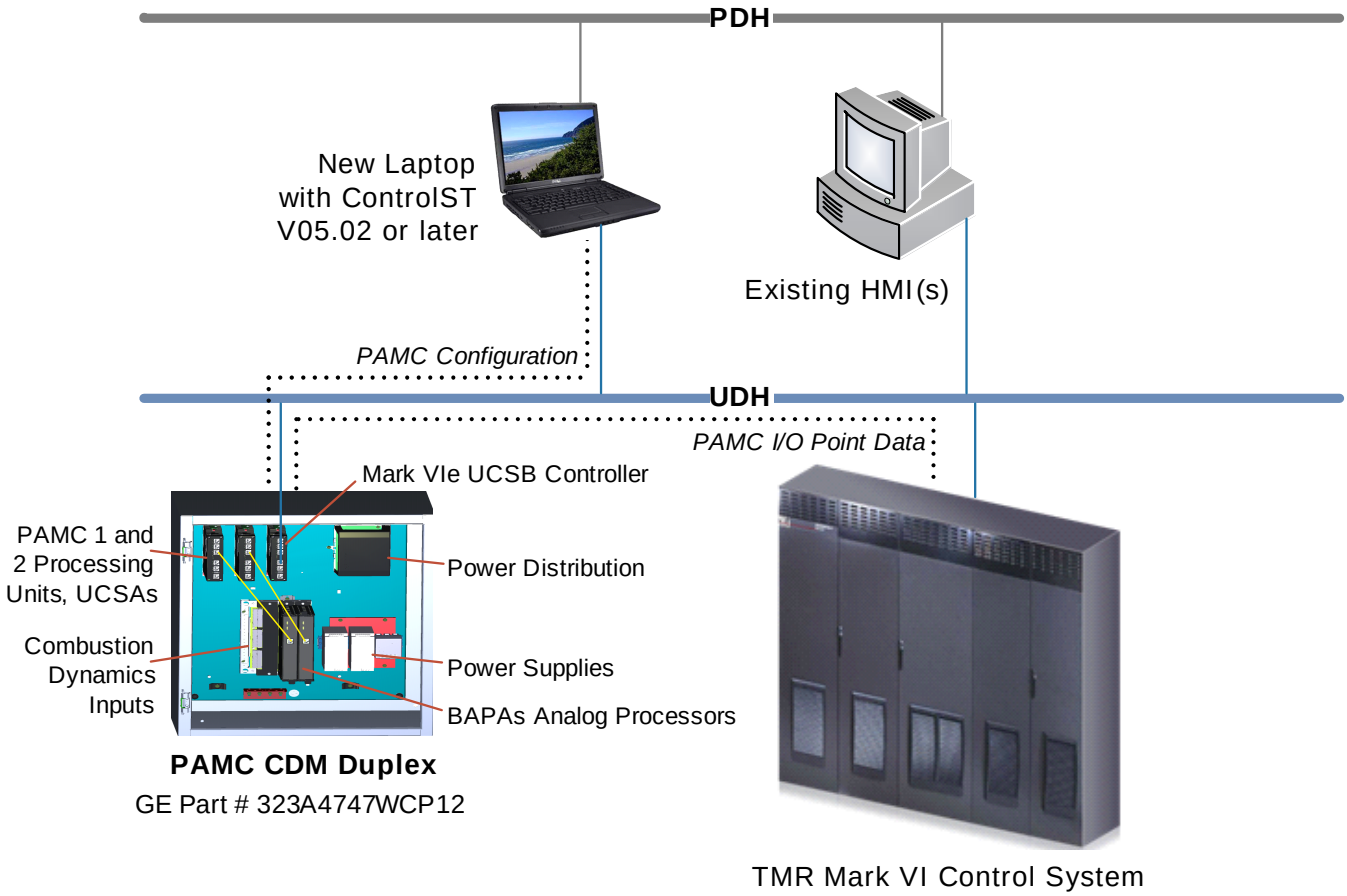
3.1 CDM Use Cases

3.1.1 Combustion Dynamics Applications for Sites with Legacy Toolbox 10 or Older

In this case, the site has Toolbox version 10 or older for Mark VI configuration tools (from an eTCSS software release), is implementing new combustion dynamics control applications, and does not want to upgrade site HMIs to Toolbox 11. The CDM solution recommended is the 323A4747WCP12 Duplex Mark VIe PAMC-based CDM (one controller, two PAMCs). The Duplex package also has redundant power supplies with input line filtering. A new computer with ControlST V05.02 or later (IC647CSTRENEW, Premium Renewables Software License Key) is required for configuration of the PAMC-based CDM. A preconfigured .tcw file is used to customize, and is then downloaded to the PAMC-based CDM after the unit has been installed into the PEECC, wired, and checked. GE P&W PGS maintains and distributes this .tcw file.

For full spectrum displays of the combustion activity, a newer eTCSS Acoustic Monitoring (AM) Gateway software (V02.06 or later) must be installed.

Since this site's configuration has Mark VI eTCSS software with Toolbox version 10 or older, the Mark VIe configuration tools cannot be used to automate the inclusion of the CDM data into the existing System Database (SDB). The Engineering technician installs a generic network device (.alb) file (legacy Toolbox configuration) to be downloaded into the Mark VI control, so that the Mark VI control can consume the EGD pages from the Mark VIe PAMC-based CDM. GE P&W PGS maintains and distributes this .alb file.



Combustion Dynamics Applications, HMIs are Older Than Toolbox 11

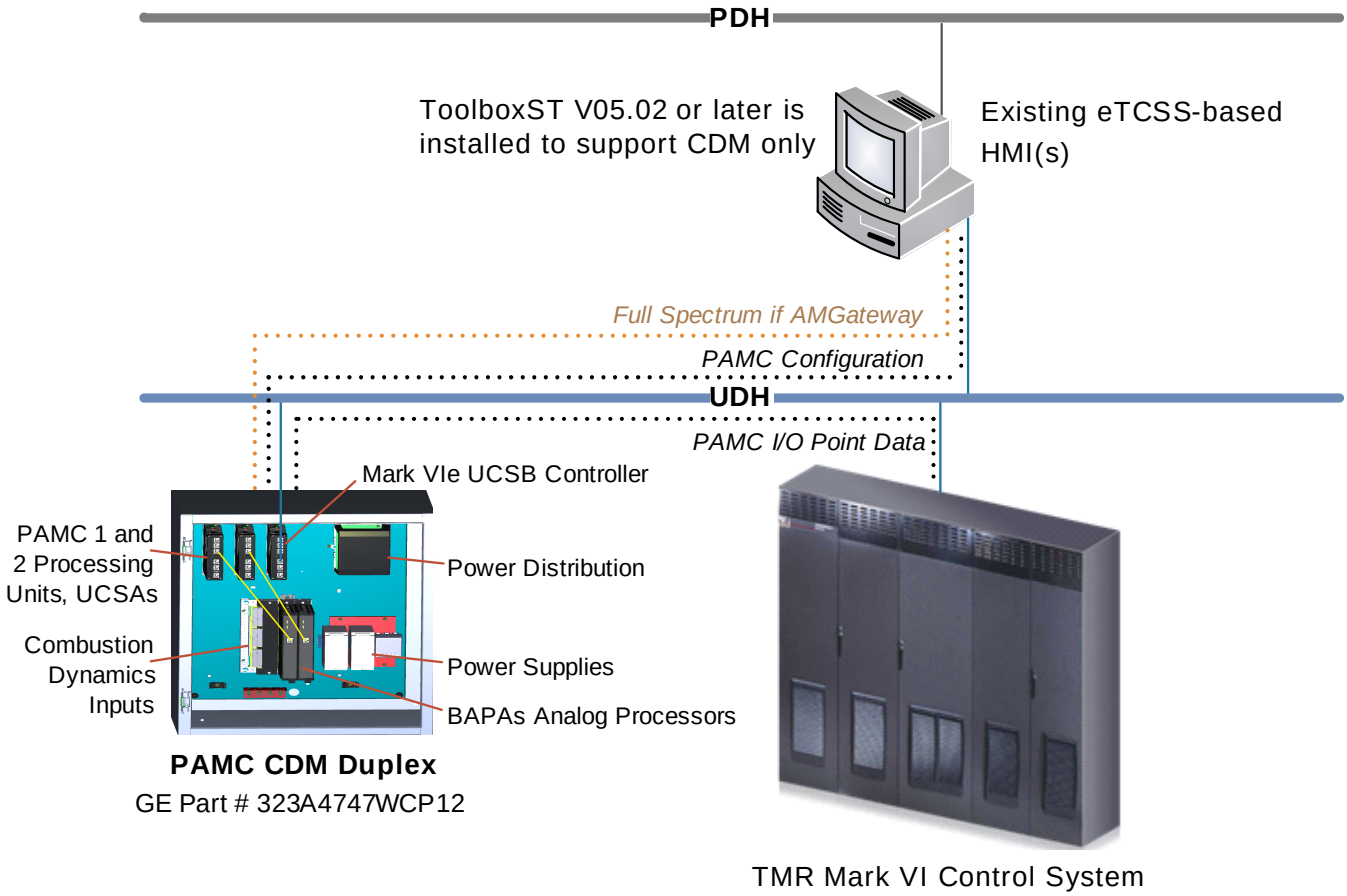
3.1.2 Combustion Dynamics Applications for Sites with Legacy Toolbox 11

In this case, the Mark VI site has Toolbox version 11 as part of their configuration tools. The CDM hardware recommended for this upgrade is the 323A4747WCP12 Duplex Mark VIe PAMC-based CDM (1 controller, 2 PAMCs). This package also includes redundant power supplies and input line filtering for each power supply. The PAMC-based CDM is enclosed in a box that is mounted to the floor of the PEECC. The field engineer can install ToolboxST V05.02 onto an existing site HMI, assuming an eTCSS-based HMI that meets the *minimum requirements* for ControlST software suite.

If the site requires full spectrum displays of the combustion activity, and if the site uses a ControlST-based HMI, then the site will need to upgrade to ControlST V05.04 or later to receive the required updates to the AMGateway.

The ToolboxST application requires a IC647CSTRENEW, Premium Renewables Software License for configuring the PAMC-based CDM hardware. A preconfigured ToolboxST .tcw file (maintained and distributed by GE P&W PGS) is used to customize, and is then download to the PAMC-based CDM after the unit has been installed in the PEECC, wired, and checked.

Toolbox version 11 will remain the Mark VI control's configuration tool, integrating the Mark VIe CDM EGD page information. In addition, Toolbox Version 11 allows sharing a common System Database (SDB) between the Mark VIe PAMC-based CDM EGD producer and the existing Mark VI control consumers of the EGD CDM data. If the site requires full spectrum displays of the combustion activity, and if the site uses eTCSS-based HMI, then the site will need the newer eTCSS AMGateway installed (V02.06 or later).



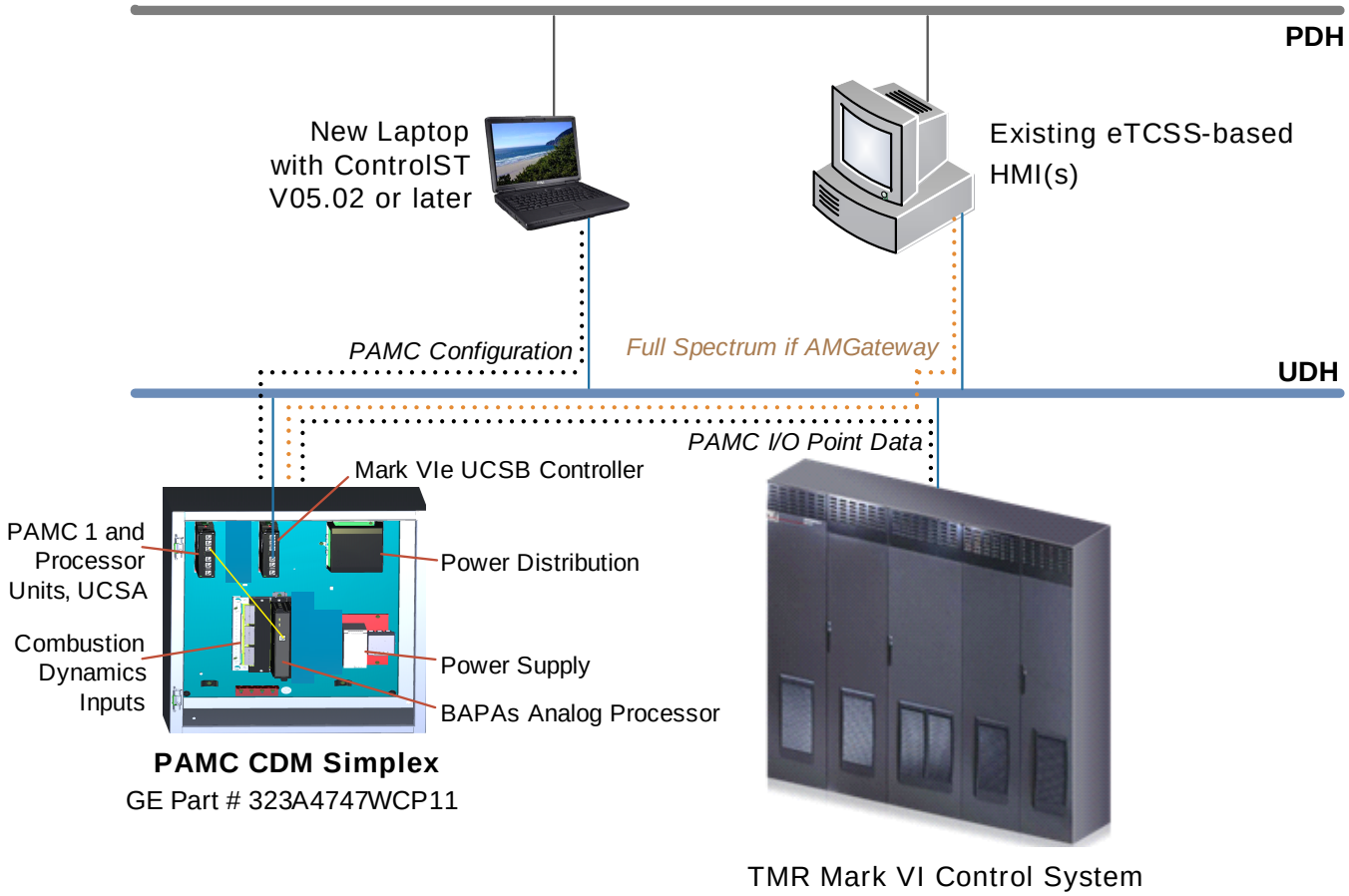
Combustion Dynamics Applications, HMIs are Toolbox 11

3.1.3 Remote DLN Tuning or CDM Only for Sites with Legacy Toolbox 10 or Older

In this case, the site has Toolbox version 10 or older for the Mark VI configuration tools based on eTCSS software, and has purchased the Remote DLN tuning service from GE or requires only CDM monitoring. The site will need the 323A4747WCP11 Mark VIe PAMC-based CDM Simplex (one controller, one PAMC). The simplex enclosure has a single power supply with input line filtering.

The site's existing eTCSS Toolbox cannot be used to configure the Mark VIe control equipment.

A new computer with ControlST V05.02 (IC647CSTRENEW, Premium Renewables Software License) is needed to configure the PAMC-based CDM hardware. A preconfigured ToolboxST .tcw file (maintained and distributed by GE P&W PGS) is used to customize, and is then download to the PAMC-based CDM after the unit has been installed and wire checked. This configuration requires no updates to the System Database (SDB) from the PAMC-based CDM to be consumed by the Mark VI control, but requires an installation of the newer eTCSS AMGateway (V02.06 or later).



Remote DLN Tuning, HMIs are Older Than Toolbox 11

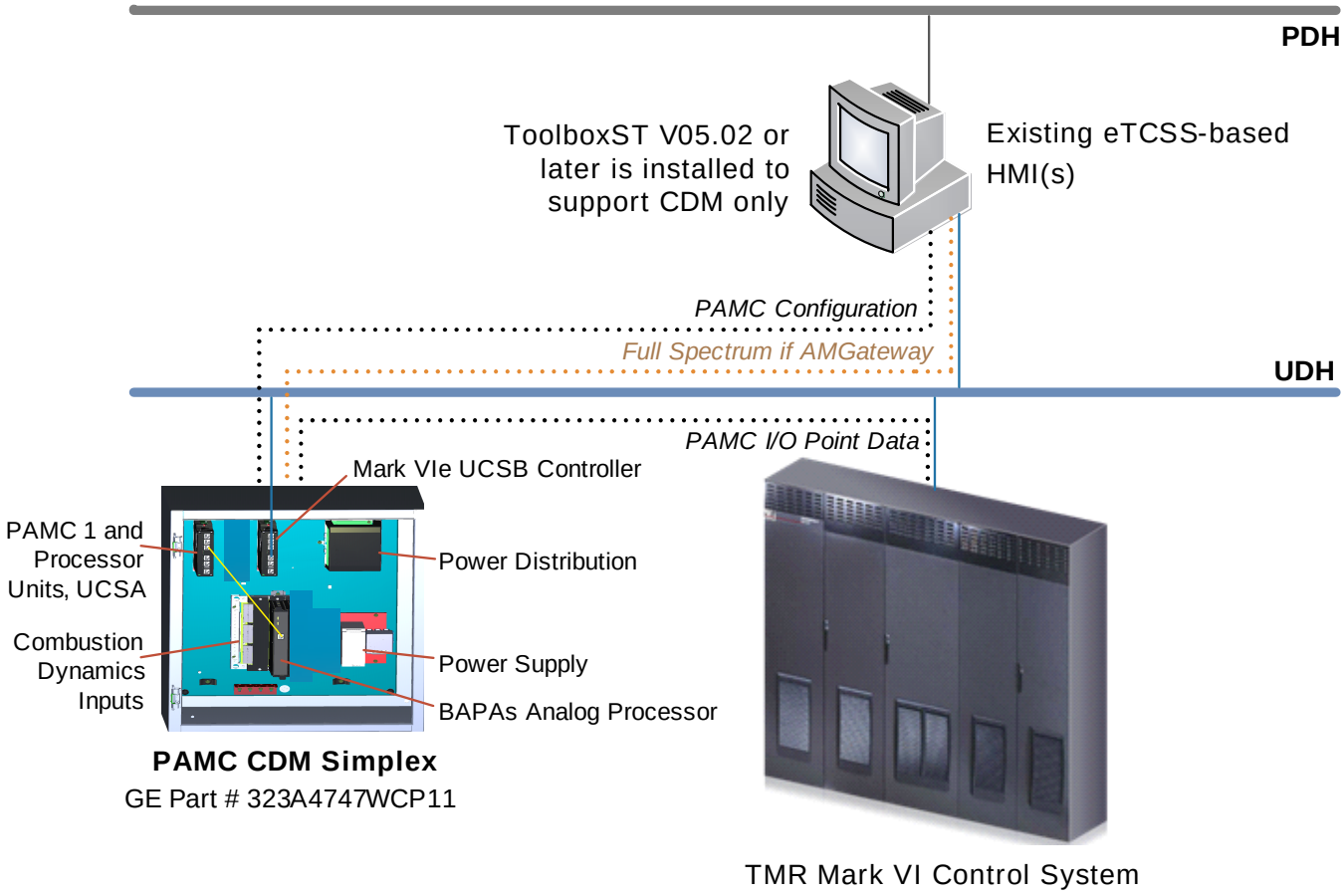
3.1.4 Remote DLN Tuning or CDM only for Sites with Toolbox 11

In this case, the site has Toolbox version 11 as part of their configuration tools. This customer has either purchased the Remote DLN Tuning service from GE or bought the CDM monitoring only product for their gas turbine. The hardware required for this option is the 323A4747WCP11 Mark VIe PAMC-based CDM Simplex (one controller, one PAMC). The simplex box includes a single power supply with input line filtering. The field engineer can install ToolboxST V05.02 onto an existing site HMI, if it is an eTCSS-based HMI that meets the *minimum requirements* for ControlST software suite.

If the site uses ControlST-based HMIs, then the site will need to upgrade to ControlST V05.04.xx using a IC647CST, full ControlST Software License.

ToolboxST V05.02 or later (IC647CSTRENEW, Premium Renewables Software License) is used to configure the PAMC-based CDM hardware. A preconfigured ToolboxST .tcw file (maintained and distributed by GE P&W PGS) is used to customize and then download to the PAMC-based CDM after the unit has been installed in the PEECC, wired, and checked. If the site uses eTCSS-based HMIs, then the site will need the newer eTCSS AMGateway installed (V02.06 or later).

Since there are no combustion dynamics control applications being applied, there are no updates to the System Database(SDB) from the PAMC-based CDM (to be consumed by the Mark VI control). Configuration of the AMGateway to support the upload of data from the PAMC to the remote DLN tuning site is required.



Remote DLN Tuning, HMIs are Toolbox 11

3.2 Orderable Parts

GE businesses can order the PAMC-based CDM by using the ENOVIA PLM tool.

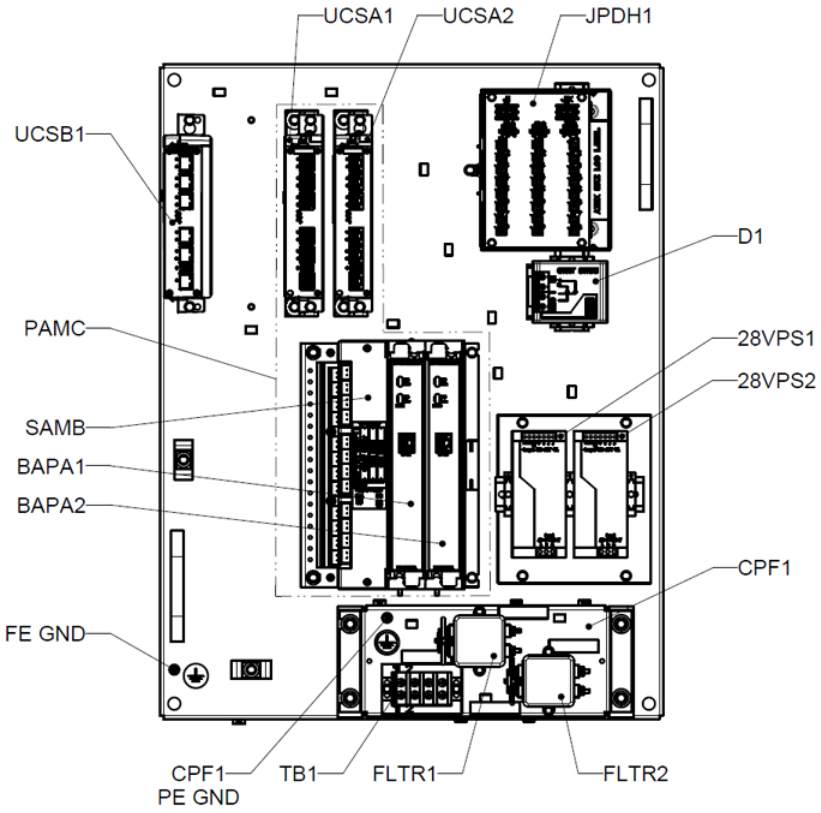
High-level Part Numbers

Part Number	Description
323A4747WCP11	Simplex: 1 UCSB controller, 1 PAMC, 1 power supply, one line filter
323A4747WCP12	Duplex: 1 UCSB controller, 2 PAMCs, and 2 power supplies, 1 diode power ORing module
114T3681G0001	Floor mounting feet (6 inches off floor) plus hardware for the junction box
IC647CSTRENEW	Premium Renewables Software License for use with ToolboxST V05.02 or later
IC647CST	Full ControlST Software License Key for use with ControlST V05.04 or later (if needing a new ControlST-based AMGateway for full spectrum displays)

Note GE Power & Water, Power Generation Services (PGS) maintains and provides the pre-configured .tcw file for the PAMC-based CDM ToolboxST configuration, and the .alb (generic network device) file for the Mark VI EGD configuration. All PAMC-based CDM ordering is managed by the GE PGS group.

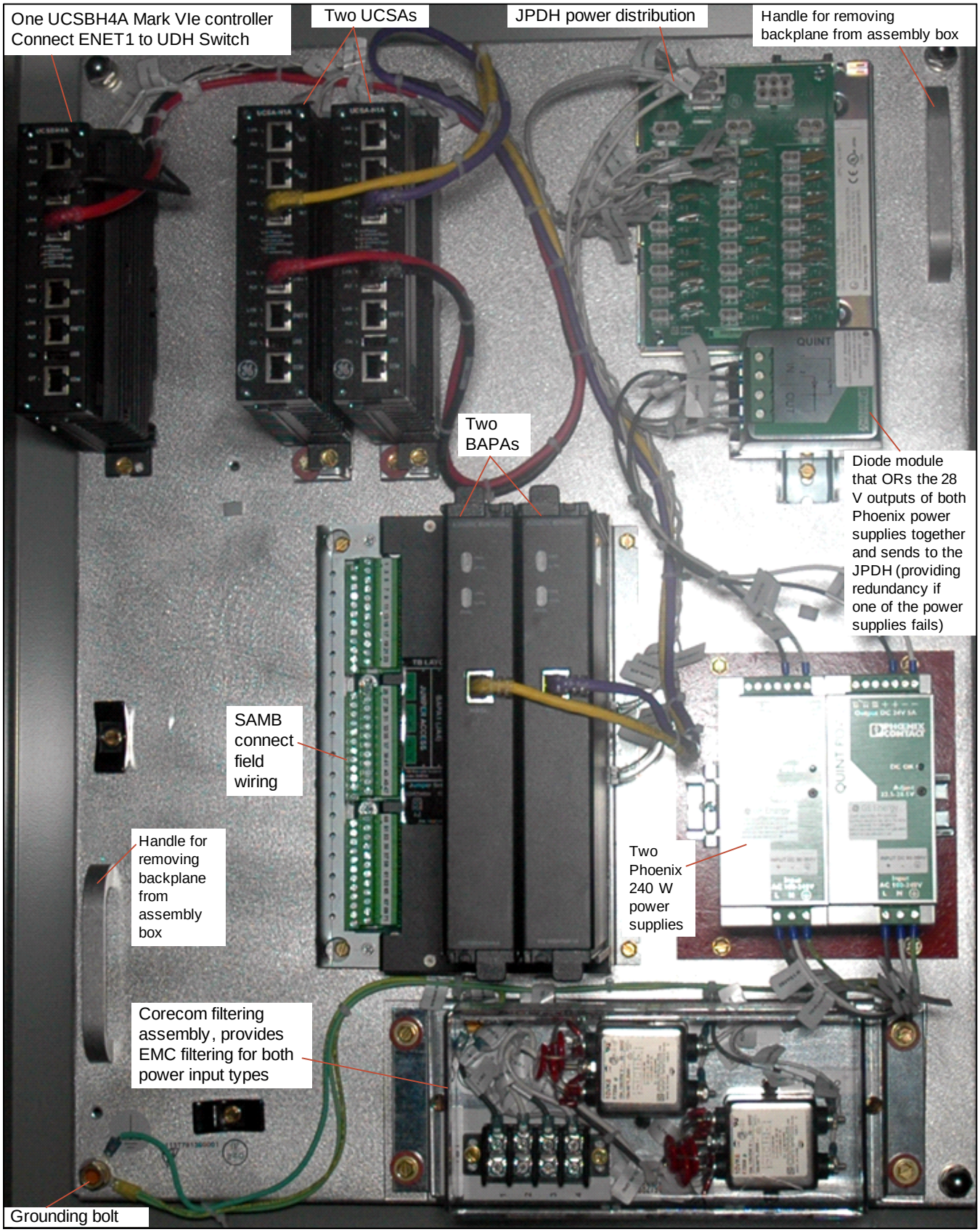
3.3 Duplex Version, Part Number 323A4747WCP12

The 323A4747WCP12 back panel includes the hardware needed to support two Mark VIe PAMCs and a UCSB controller. This assembly can be powered by either a 125 V dc or 110/220 V ac source.



323A4747WCP12 Back Panel – Duplex Version

Located at the bottom right portion of the back panel assembly is a Corecom filtering assembly, CPF1. This provides EMC filtering for either ac or dc power input types, which are connected to TB1. The Phoenix power supplies (28VPS1 and 28VPS2) convert the ac or dc input power to 28 V dc power. The redundant 28 V dc power runs through a diode module that provides power ORing, and then JPDH1 distributes the power to the PAMCs (UCSAs), the UCSB controller, and the BAPAs.



Duplex Version

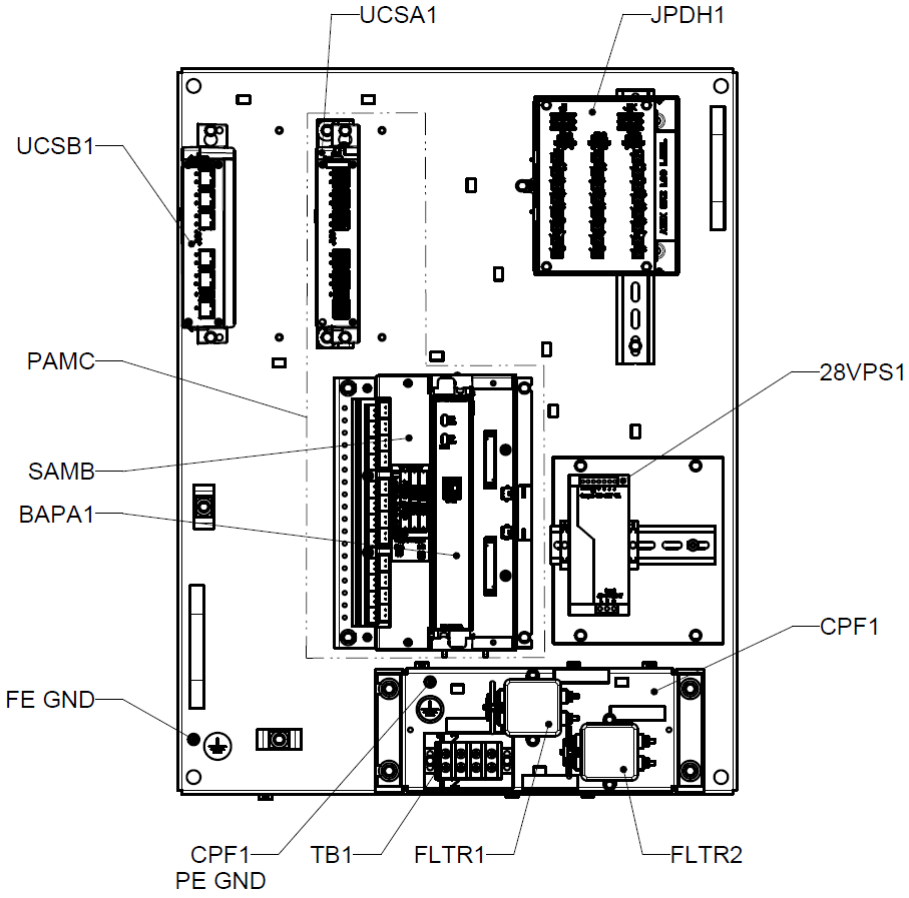
3.4 Simplex Version, Part Number 323A4747WCP11

The simplex box can be upgraded to duplex by ordering and installing the additional, UCSA, BAPA, Phoenix power supply, and diode power ORing module.

The 323A4747WCP11 back panel includes the hardware needed to support one PAMC module as follows:

- One UCSA
- One UCSB
- One BAPA
- One Phoenix power supply
- One Corecom filtering assembly

This simplex PAMC-based CDM can be powered by either a 125 V dc or 110/220 V ac source. On the Corecom input filter, connect the source to one set of screws on TB1 corresponding to a single source feed.



Simplex Assembly

3.5 Summary of Hardware Installation

The assembly may be wall mounted or floor mounted. The floor-mounting option requires the hardware legs (part number 114T3681G0001).

The installation scope is comprised of the following:

- Detach the door and remove the back panel.
- Attach part # 114T3681G0001 legs to the metal box and mount it to the PEECC floor.
- Reattach the back panel. Take care in reattachment of the bolt that has the grounding strap with the ring terminal connections.
- Attach conduit interface to box, run wiring through conduit and connect to the terminal block on the Corecom input filter for power and connect the ground to the bonded 1/4" x 20 stud on the left sidewall of the junction box.
- Attach conduit interface to box, run Ethernet cable for UDH and connect to the UCSBH4A controller ENET1 port.
- Attach conduit interface to box, run wires from charge amplifier box to connect to the SAMB terminal board.
- Reattach the door.

3.6 Mounting

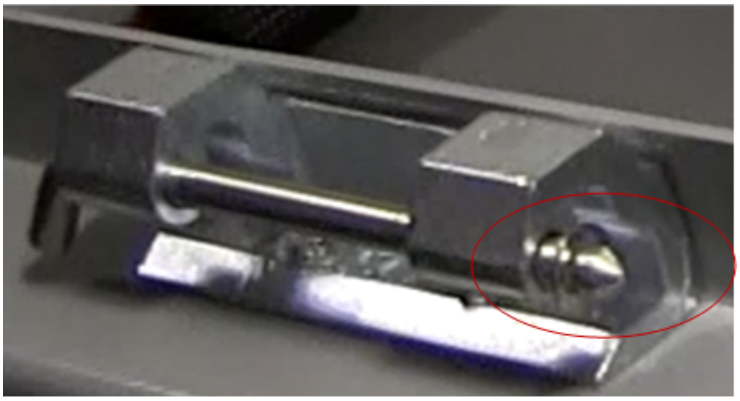
The following table provides specifications for the enclosure and mounting hardware.

Item	Specification
Dimensions	Inches
Door Swing	Clearance required at front is 24 inches 180 degrees maximum swing Removable by disengaging the lift-off hinge pins in the concealed hinges
Door locking hasp and staple	Use standard 3/8” lock or less in lock shackle
Wall mounting	Use screws at all four designated mounting points Grade 12.9: ½” HEX socket cap screws (IISO 4762 DIN 912/JIS B1176 or similar) Remove door and back panel for less weight during installation
Floor mounting kit	Required legs: GE part # 114T3681G0001
Enclosure rating	IS6 (NEMA Type 4, 12 & 13) rated – outdoor use
Enclosure materials	Walls and door: 0.075 inch carbon steel Mounting tabs: 0.149 inch carbon steel Finish is ANSI-61 gray powder coating
Weight	Total junction box: 95 pounds (43.09 kg) Removable door: 17 pounds (7.71 kg) Removable electronics on back-panel: 47 pounds (21.32 kg) Empty junction box without door: 46 pounds (20.87 kg)
Approximate total max case power requirements	75 watts
Width of opening for cabling	28 inches
Bottom entry for cabling	Center the unit in the opening leaving approximately two inches on each side
Side entry for cabling	Add whatever additional width is required for the customer conduit

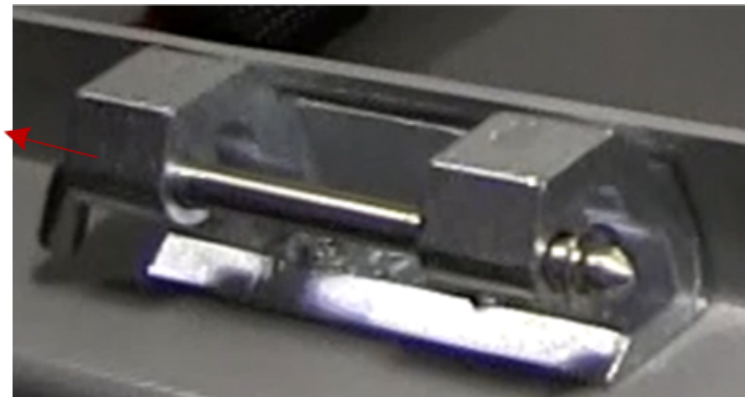
3.6.1 Detach the Door and Remove the Back Panel

➢ **To remove the door.**

1. Locate the two removable hinges.



2. Remove the C-clip from each hinge.



3. Slide the pin out from each hinge.

4. Lift the door off and carefully set it aside.

➢ **To remove the back panel.**

1. Locate the four bolts and remove them.



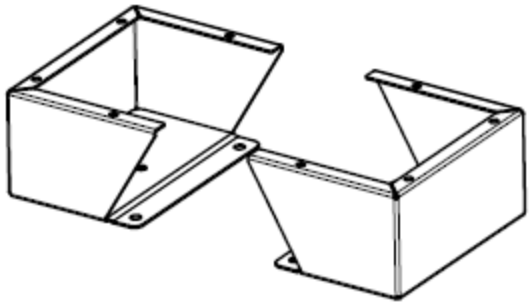
2. With the bolt that has the grounding strap, be sure to keep the loose ring terminal connections with the back panel for reattachment later.
3. Use the handles to carefully lift the back panel and set it aside.



3.6.2 Floor Mounting

Be aware that the distance from the Mark VI enclosure to the PAMC-based CDM may require additional wire lengths.

The floor mounting feet will raise the junction box 6 inches off the floor. These feet are attached to the bottom of the assembly and secured to the PEECC floor.



Floor Mounting Feet, Part Number 114T3681G0001

3.6.3 Wall Mounting

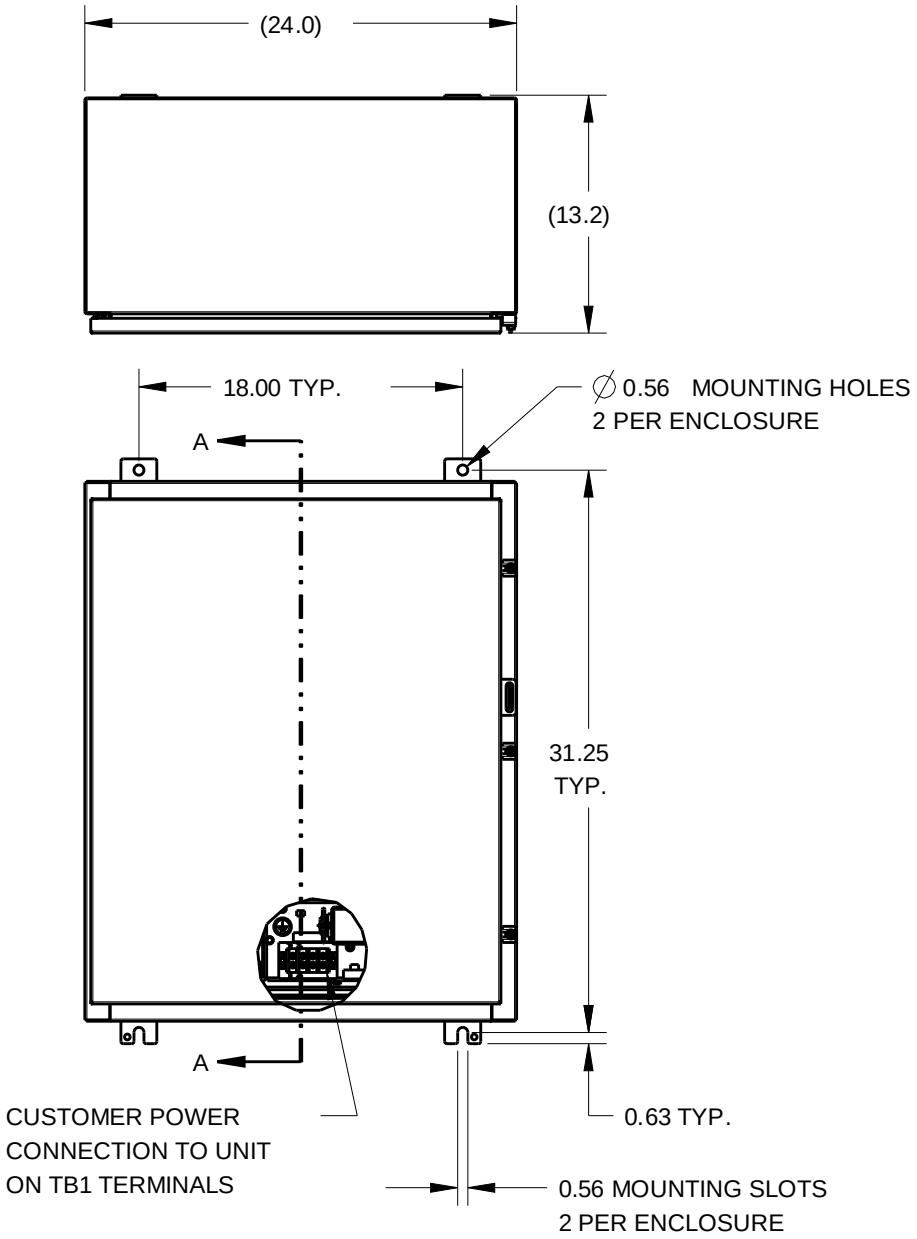
If the unit must be wall mounted (rather than floor mounted), it is attached to wall studs or equivalent wall structural members. the unit should be attached using four, 1/2" diameter Grade 5 steel fasteners. If vertical studs or equivalent structure is not accessible, then the box may be mounted to horizontal supports which are secured to wall studs or equivalent structure. The horizontal supports must be adjusted to support 200 pounds (90.72 kg), distributed between the mounting points attached to the support.

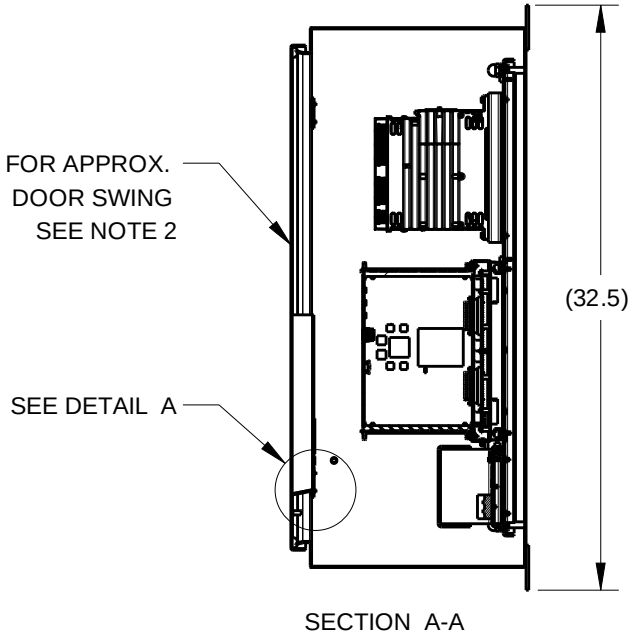


Attention

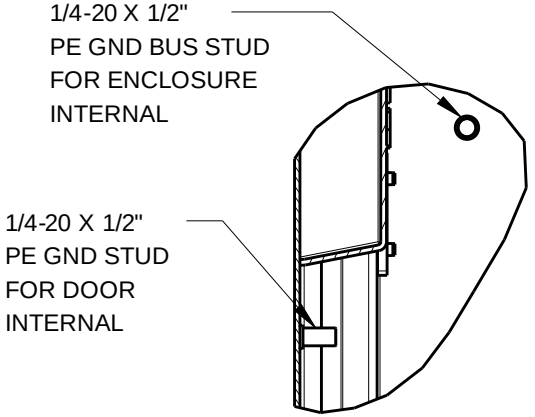
The unit should not be directly mounted to drywall or other similar low-strength wall materials.

3.6.4 Mechanical Drawing





SECTION A-A



DETAIL A
SCALE 0.500

3.7 Wiring and Connections

3.7.1 Input Power and Protective Earth (PE) Grounding

The 125 V dc source or a 110/220 V ac source input power is connected to the TB1 terminal strip on the Corecom filter, located at the bottom of the junction box.

For duplex PAMC-based CDM, two ac or dc inputs are provided, allowing redundancy in power. Preferred method is 110/220 V ac source. On the Corecom input filter, connect the source to both sets of screws on TB1 corresponding to the correct source feed.

The simplex PAMC-based CDM can be powered by either a 125 V dc or 110/220 V ac source. On the Corecom input filter, connect the source to one set of screws on TB1 corresponding to a single source feed.



Warning

To prevent shock, follow proper Lock Out Tag Out (LOTO) procedures before connecting power to the TB1 connection strip on the filter assembly located at the bottom of junction box.

Customer Input Ratings

Item	Specification
Voltage	AC: 58 to 264 V ac nominal, 45 to 65 Hz
Power	120 W
Terminal Designations	Input1-Line: TB1-1, Input1-Neutral: TB1-2 Input2-Line: TB1-3, Input2-Neutral: TB1-4
Terminal Ratings	550 V, 28 A (EN 60079-7)
Terminal Specifications	Spade Shape; Stud Size: #8
Connecting Wire Size	AWG 14
Connecting Wire Insulation Rating	105ºC, 600 V minimum
Suggested Breaker Size	AC: 5 A DC: 10 A
Protective Earth Connection	Minimum Wire Size: AWG 8 stranded to be connected to a 1/4" x 20 bonded stud in the junction box

3.7.2 UCSB to UDH

Connect the Ethernet cable from the ENET1 port of the UCSB controller to the network switch UDH port using the following guidelines:

There have been field issues with AT8026 to Mark VIe controller communications.

- Verify that the UDH switch is 100 MB speed.
- If the site currently uses Allied Telesis 8026 switches for UDH, GEIP recommends that the site upgrade to the new Cisco 2960S switch for UDH connection to the Mark VIe controller.
- The Ethernet cable is Cat 5e UTP.
- The Cat 5e cable maximum distance is 100 meters (328.08 feet).

3.7.3 Inputs to the SAMB

Refer to GEH-6721_Vol_III, the chapter PAMC Acoustic Monitoring Input Module for the information on field wire connections to the SAMB.

Possible wiring conditions:

- If site has an existing VAMB:
  - If possible, move the wiring from TAMB terminal board to SAMB terminal board.
  - Install the PAMC-based CDM junction box as close to Mark VI cabinet as possible.
- If site is replacing a previous EdasCE system then there is a junction box located outside the Mark VI enclosure. Remove the EdasCE and reconnect wiring to PAMC-based CDM, which is installed in the same place that the EdasCE was located.
- If no EdasCE is present, locate the PAMC-based CDM box anywhere in PEECC, preferably a minimum 6 to 12 inches away from any limit amp cabinet.

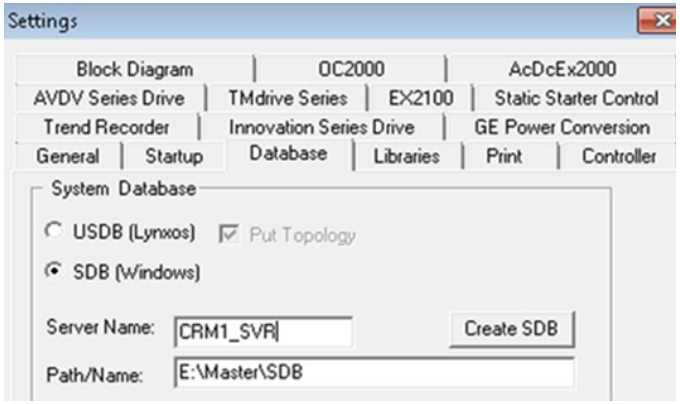
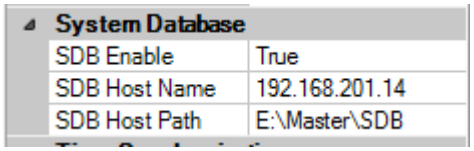
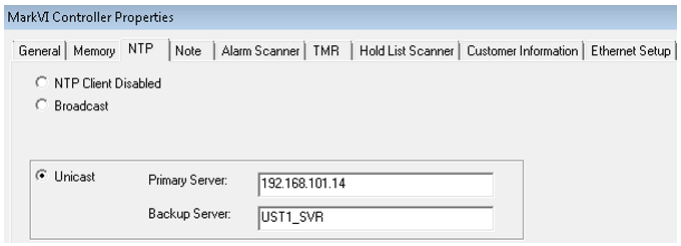
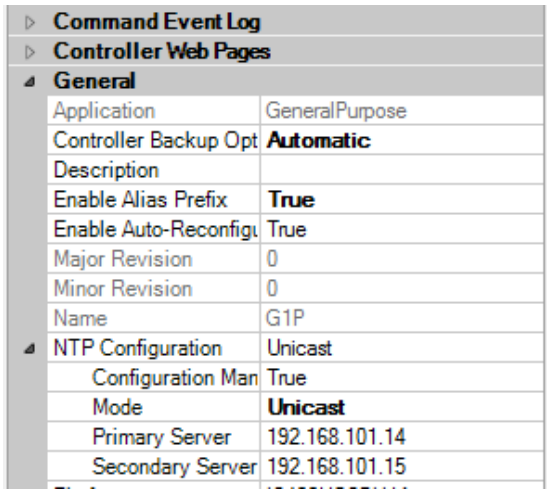
Connect to the terminal screws on the SAMB, the signals that come from the output of the charge amplifiers. The PAMC-based CDM can interface with Encore or Gen IV CDM charge amplifier / probe systems.

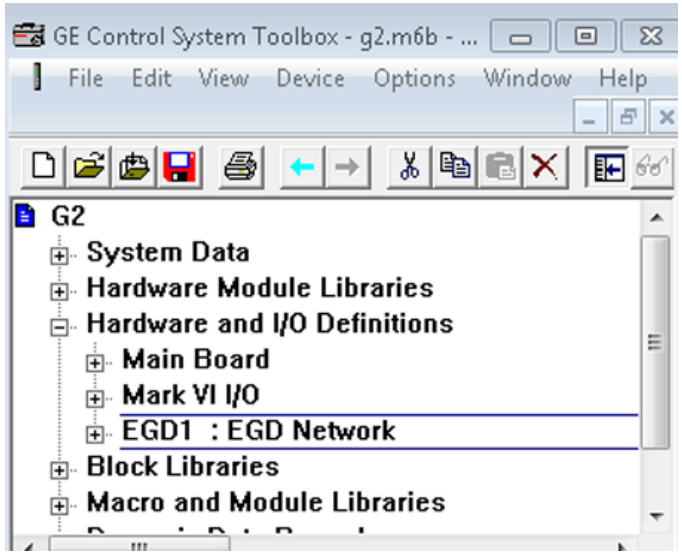
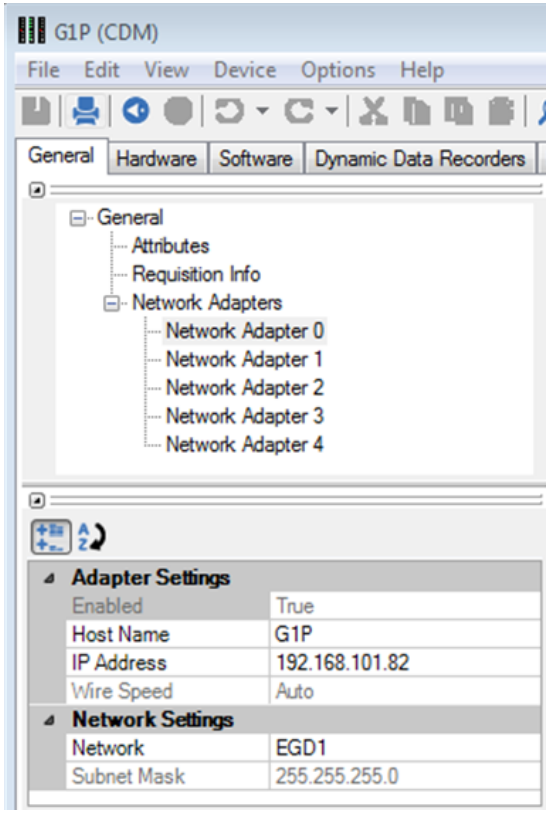
The Encore JB1000 junction box contains the ENC 1492A charge amplifier boards (inside). This is a 110/220 V ac externally powered 18-channel charge amplifier system. The ENC 1492A accepts inputs from either the Endevco or PCB probes. The probe charge output (pico-coulombs) is converted to a milli-volt output that is wired to the SAMB terminal board.

The Gen IV PCB Piezotronics Model 682M57 or EX682M75 charge amplifier does the same thing as the Encore device, but instead of being powered externally by a 110/220 V ac source, the device is powered by a 3.6 mA constant-current source supplied by the SAMB on the same two wires, providing the combustion dynamics signal in millivolts. The Gen IV eliminates the external 110/220 V ac supply and also provides a nominal 12 V dc bias voltage monitored by the PAMC, providing improved diagnostics over the Encore system.

4 Appendix

The following table is provided for the field engineer to help with comparing the configurations of the classic Toolbox and the ToolboxST applications.

Item	Location in Toolbox	Enter Value Here	Location in ToolboxST
SDB Server (typically 192.168.201.14)			
Path to SDB (typically E:\Master\SDB)			
Mark VI NTP Server			

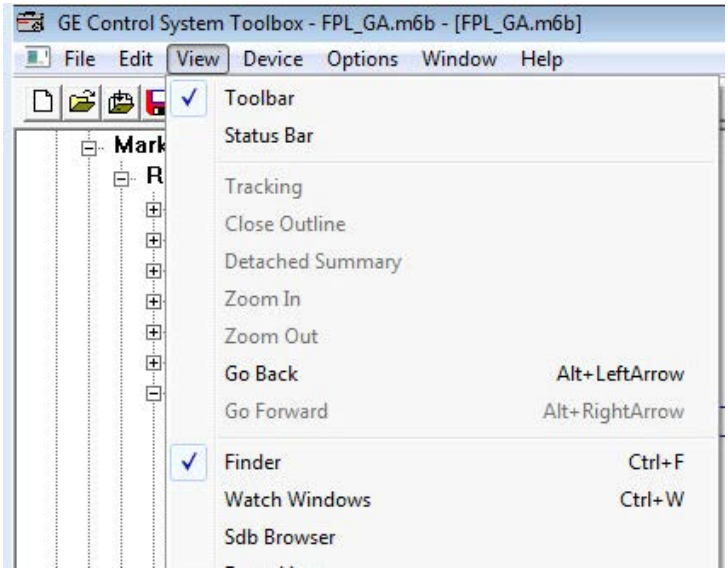
Item	Location in Toolbox	Enter Value Here	Location in ToolboxST
EGD Network Name			

4.1 Undo a Put to SDB

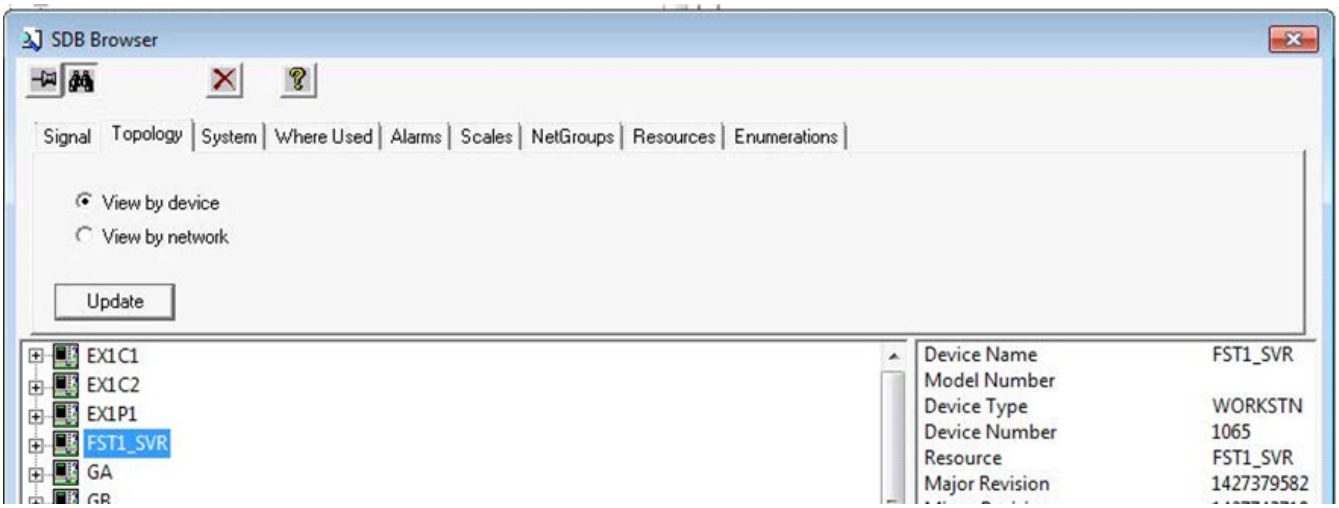
This procedure would be used if someone accidentally puts a WorkstationST device into the Mark VI SDB.

➢ **To undo a put to the Mark VI SDB**

1. From the classic Toolbox application, select **View**, **Sdb Browser**.



2. The SDB Browser displays, click **Update**.
3. The devices in the SDB populate. Select the device to be removed from SDB, then click the red "X" at the top of the SDB Browser.



4. A Toolbox Warning dialog box confirms that you want to delete the device from the SDB. Click **Yes**. The device has now been deleted. This can be confirmed by viewing the SDB Browser again.